Public Information, Private Expertise, and Lender Specialization

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Abstract

Why do lenders specialize in specific industries rather than diversify? Exploiting the SBA's 2014 credit-scoring mandate, which applied to most guaranteed loans but left one program exempt, I show that standardized scores reduced specialization by 12.7 percent as lenders entered new sectors and reallocated portfolios. Yet public information crowds out private expertise: in previously specialized industries, post-reform defaults rise by 11 percent of the pre-policy rate while competitive entry holds spreads flat. A Bartik shift-share IV confirms that remaining specialization causally lowers defaults, deepens repeat borrowing, and reduces spreads for repeat borrowers, reflecting informational complementarities rather than hold-up.

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Despite the rule of thumb favoring diversification in lending (Diamond, 1984), many lenders concentrate their portfolios in a narrow set of industries, reflecting persistent specialization rather than broad risk spreading. This concentration embodies an information-acquisition choice: by specializing, lenders develop industry expertise that improves screening and monitoring. But how valuable is this expertise, and what happens when a new source of public information reduces screening costs and narrows information gaps across lenders?

Recent empirical studies document that industry-specific expertise is associated with superior lending outcomes (Blickle et al., 2023; Cabossioras and Tielens, 2024; De Jonghe et al., 2025; Giometti et al., 2024), but this evidence is largely correlational and focused on within-industry effects.[1] On the theoretical side, Blickle et al. (2025) provide a foundation, showing that lenders develop persistent specialization and use private signals to screen borrowers, achieving lower defaults and lower interest rates. This paper complements their framework with the first causal evidence that a public-information shock reshapes both the level and the value of lender specialization. Exploiting a 2014 SBA credit-scoring mandate that exogenously expanded public information, I show that public information substitutes for private expertise, reducing specialization and improving average screening; but wherever specialization persists, it provides lenders with lower defaults, stronger borrower relationships, and more favorable pricing, confirming its role as a durable informational advantage.

On the extensive margin, the credit-scoring mandate reduced lenders' industry specialization by 12.7 percent relative to the pre-reform average. Decomposing this decline, I find that treated lenders both entered industries in which they previously did not originate loans and reallocated credit away from pre-policy overweights toward the market benchmark, as standardized information lowers barriers to entry and erodes the rents that sustain concentrated portfolios.

However, this expansion comes at a cost. While average default rates fell by approximately 1.0 percentage point, reflecting more precise screening, the gains are unevenly distributed. In industries where lenders were previously most specialized, public information displaces private screening expertise: a one-standard-deviation increase in prior specialization raises post-reform defaults by 11 percent of the pre-policy default rate, partially offsetting the baseline improvement. This result is reinforced by the

[1] By contrast, Paravisini et al. (2023) study Peruvian exporters, Hastings et al. (2017) examine Mexican pensions, Benetton (2021) analyze U.K. mortgages, and Duquerroy et al. (2022) investigate French local banks.

screening channel: the policy's default reduction concentrates among new borrowers lacking prior relationships. The dark side of public information is that standardized scores cannot replicate the industry-specific knowledge that specialized lenders accumulated through repeated lending, so the resulting loss of screening quality is largest for the borrowers who benefited most from specialized lenders.

On the intensive margin, I address endogenous loan allocation using a Bartik-style shift-share instrument that holds each lender's pre-2014 industry credit shares fixed and interacts them with national changes in industry loan demand. The instrumental-variables estimates show that specialized loans have default rates 1.2 percentage points lower (21 percent of the pre-policy average), are 6.5 percentage points more likely to involve relationship lending (43 percent of the pre-policy rate), and carry spreads 22.4 basis points lower for repeat borrowers. The default and relationship effects reflect screening and monitoring advantages from industry expertise; the pricing effect for repeat borrowers reflects informational complementarities between industry expertise and borrower-specific knowledge rather than rent extraction, supporting the view that specialization enhances screening precision rather than enabling hold-up (Rajan, 1992). These gains vary across industries: specialization reduces defaults in sectors with verifiable collateral, where expertise aids asset valuation, but concentrates correlated exposure in demand-driven sectors.

The identification exploits a July 2014 SBA policy that required lenders to screen Standard Guaranty loans up to $350,000 using a standardized credit score (SBSS), while Express loans—which follow lenders' own credit policies—remained exempt. This differential exposure creates a natural experiment: Standard loans form the treatment group and Express loans the control. To measure specialization, I build on the normalized revealed comparative advantage framework of Yu et al. (2009) and construct a novel, scale-invariant index that sums to zero across both lenders and industries, permitting consistent cross-lender and cross-industry comparisons, and captures zero-origination decisions, which is critical for measuring entry into new sectors.

This paper contributes to three strands of literature. The first is lender specialization. Theory predicts a tradeoff between diversification, which reduces failure risk (Diamond, 1984), and concentration, which dominates when information rents or monitoring incentives are strong (Acharya et al., 2006; Winton, 1999). Recent empirical work documents that banks systematically specialize by industry and that this specialization improves lending outcomes: Blickle et al. (2023) show that specialized banks

have lower charge-off rates and lend more to opaque firms; Di and Pattison (2023) find that industry-specialized SBA lenders achieve better loan performance at distance; and De Jonghe et al. (2025) demonstrate that specialized banks are less likely to extend credit to zombie firms. Related work links specialization to contract design (Giometti et al., 2024), corporate innovation (Degryse et al., 2025), and monetary policy transmission (Casado and Martinez-Miera, 2024). Whereas existing studies treat specialization as a fixed lender characteristic, I show that an exogenous reduction in information asymmetry shifts the equilibrium level of specialization, and introduce the extensive margin of specialization adjustment, decomposing portfolio responses into entry and reallocation across industries. The Bartik-style instrument further provides causal identification of specialization's value within a single institutional framework, complementing the correlational evidence in Blickle et al. (2023) and the cross-country settings of Paravisini et al. (2023) and Duquerroy et al. (2022).

The second contribution is to information production in lending. A central distinction in this literature is between hard information, which can be credibly transmitted through hierarchies, and soft information, which is produced through relationships (Berger et al., 2005; Stein, 2002). Technological change has expanded the role of hard information: Petersen and Rajan (2002) document that information technology increased the distance between small firms and their lenders, while credit scoring has altered who receives credit and on what terms (Einav et al., 2013; Frame et al., 2001; Frame and Woosley, 2004). Blickle et al. (2025) formalize how specialized lenders exploit private signals to price more competitively, and He et al. (2023) show that data sharing can erode incumbent banks' information rents—but no direct empirical test of these predictions exists. The SBA reform provides one: an exogenous hard-information shock that documents how standardized signals substitute for industry expertise (Hauswald and Marquez, 2003), with asymmetric consequences for performance and pricing that confirm the theoretical prediction.

Third, this paper contributes to the relationship lending literature. Lending relationships generate value through repeated interaction (Berger and Udell, 1995; Petersen and Rajan, 1994), geographic proximity confers informational advantages in pricing (Degryse and Ongena, 2005), and concentrated markets foster relationship investment (Petersen and Rajan, 1995), but Rajan (1992) shows that informed lending can also create hold-up power. Boot (2000) emphasizes that relationship banking involves proprietary information reused across interactions, and Cabossioras and Tielens (2024) show that industry-specialized lenders offer lower initial rates to relationship

borrowers. A key question these studies leave open is whether industry expertise and borrower-specific knowledge are substitutes or complements, and whether the resulting pricing advantages reflect lower costs or hold-up. The IV estimates resolve both: specialization causally increases repeat borrowing and lowers spreads for returning borrowers, reflecting informational complementarities rather than rent extraction.

The remainder of the paper is organized as follows. Section 1 describes the SBA 7(a) program. Section 2 develops the conceptual hypotheses. Section 3 explains the specialization measure and describes the data. Section 4 analyzes specialization on the extensive margin in the SBA market. Section 5 examines the effects of specialization on the intensive margin. Section 6 concludes with policy implications and directions for future research.

1 Institutional Background

1.1 SBA 7(a) Loan Program

The 7(a) loan program is the Small Business Administration's (SBA) largest guarantee initiative, dating back to 1953. Under the program, private lenders extend credit to small businesses, while the SBA reviews each application and backs approved loans with a federal guarantee. By 2019, annual 7(a) originations had reached $23 billion, roughly three times the volume of the next-largest SBA program. Loans carry one of three typical guarantee levels (50, 75, or 85%) and are further differentiated by maturity and maximum loan size.[2]

7(a) loans can be grouped into two broad subprograms. First, *Standard Guaranty* loans, which receive a 75–85% guarantee, are split into small loans (principal of $350,000 or less) and regular loans (principal above $350,000). Second, *Express* loans carry a 50% guarantee and cannot exceed $350,000.[3] Table 1 summarizes key program terms—loan size, guarantee levels, maturity, rate spreads, and fees—for 7(a) Small and Express loans. A central feature of SBA lending for this study is that borrowers pose elevated default risk. Eligibility is restricted by the "credit not available elsewhere" requirement, which means borrowers must first demonstrate an inability to

[2]Certain Export Working Capital loans and lines of credit have a 90% guarantee.

[3]Because the guarantee share is higher in the standard program, SBA scrutiny is correspondingly tighter. Preferred Lenders Program (PLP) institutions, however, face an abbreviated review of about seven days and may self-underwrite within SBA guidelines.

secure comparable financing on conventional terms. Firms, therefore, often lack sufficient collateral, have short operating histories, or need longer maturities than banks ordinarily provide. These characteristics make SBA applicants relatively opaque and risky, heightening information asymmetries for lenders.

To mitigate this opacity, the SBA guarantees a fixed share of each loan. If a borrower defaults, the agency reimburses the lender for that share.[4] Even with the guarantee in place, lenders retain strong incentives to screen and monitor. The SBA designates high-performing institutions as Preferred Lenders, granting them authority to process, close, service, and liquidate most guaranteed loans without prior SBA review. Because qualification depends on satisfactory default and loss histories, active SBA lenders must maintain rigorous screening and monitoring to preserve their Preferred Lender status.

Collectively, these institutional features create an environment in which private information and industry expertise play a central role in credit allocation, making the 2014 credit-scoring reform a particularly informative shock.

1.2 Information Shock: SBA Policy Change in 2014

In January 2014, the SBA introduced a significant underwriting change for standard 7(a) guarantees up to $350,000 (Express loans remained exempt). Lenders must now transmit borrower data through the Electronic Loan Processing/Servicing (E-Tran) system, which generates a FICO Small Business Scoring Service (SBSS) score derived from consumer credit records, commercial bureau data, borrower financial information, and application details. Loan approval hinges on that score: applications falling below the threshold are automatically declined. The initial cutoff, set in July 2014, was 140 on the 0–300 scale.[5]

The policy matters for two reasons. First, it narrows the information gap among lenders. The new screening tool provides every 7(a) lender, including small community banks, with a low-cost credit score, reducing the advantage held by large national lenders that already employ sophisticated models. Second, the SBSS combines several data sources into a single metric, yielding a cleaner and less noisy signal of borrower quality than many proprietary systems. The reform therefore introduces a standard,

[4]The guarantee covers the charged-off balance, typically between 50 and 85%. Lenders may sell the guaranteed portion of term loans on the secondary market, retaining servicing responsibilities.

[5]On 28 August 2020, the SBA raised the cutoff to 155, periodically adjusting the threshold to track the lower tail of its risk distribution. Because the sample period ends in Q1 2020, this adjustment falls outside the analysis window.

publicly available credit measure that enhances screening precision across the SBA network.

The reform has a direct implication for lenders' information-acquisition incentives. Before the policy, lenders that specialized in particular industries accumulated private, industry-specific knowledge that yielded an informational advantage over less specialized competitors. The SBSS score compresses this advantage by providing all lenders with a common baseline signal at zero cost, reducing the marginal return to costly private information acquisition and the incentive to concentrate lending in industries where lenders have built expertise. When screening at origination becomes cheaper and more precise, the relative return to monitoring through ongoing relationships also declines: lenders can rely more on hard information at origination and less on soft information accumulated through repeated interactions.

While the new credit scoring took effect for most loans, Express program loans were not subject to the same change. Express loans follow lenders' own credit policies and do not require SBA scoring.[6] This exogenous policy shift creates a natural experiment: loans subjected to the new screening (all but Express) form the treatment group, while Express loans serve as the control group. Table 2 summarizes the 2014 screening changes.

Although borrowers seeking loans below $350,000 could, in principle, apply under either the Standard Guaranty or the Express program, economically meaningful substitution between programs is likely limited. First, the guarantee structure makes steering marginal borrowers into Express unattractive. Express loans carry a 50% guarantee, compared with 75–85% under the Standard program, so lenders retain roughly twice as much default exposure in Express. For applicants near the new screening threshold, shifting into Express would therefore increase expected losses unless the lender has strong private information indicating that the credit score understates borrower quality. Second, SBA interest rate caps constrain repricing in both programs, limiting lenders' ability to offset the higher retained risk in Express through higher interest rates. SBA operating procedures also require Express lenders to apply the same credit standards used for similarly sized non-SBA commercial loans, reducing scope to use Express as a channel to bypass the new screening regime.

Figure 1 plots mean log lending volume for the treated Standard program and the control Express program, normalized to the reform quarter (2014:Q3). The two series

[6]The standard operating procedures state that lenders that do not use credit scoring for their similarly sized non-SBA commercial loans may not use credit scoring for SBA Express or Export Express loans.

track each other closely before the 2014 policy change, confirming similar pre-trends across groups. After implementation, lending volume remains relatively flat in the control group, while the treated group shows sustained growth. The flat post-reform trajectory of Express volume mitigates concerns about strategic substitution: if lenders were steering borrowers into Express to avoid the new screening mandate, Express originations should rise, not remain stable.

Although Standard and Express loans differ in their guarantee rates (75–85% vs. 50%), typical maturities, and product mix, these level differences do not threaten the identification strategy. The difference-in-differences design absorbs all time-invariant differences through lender-industry-loan-type fixed effects; identification requires only that *trends* in specialization would have evolved in parallel absent the reform. Several features support this assumption. First, both programs draw from the same pool of SBA-eligible small businesses and are originated by many of the same lenders—the comparison exploits within-lender variation across subprograms, not cross-lender differences. Second, the formal CSDID pre-trend tests fail to reject within the eight-quarter pre-reform window ($p = 0.152$ for $|Spec|$, $p = 0.457$ for $Spec^2$, $p = 0.838$ for HHI), confirming that specialization patterns moved together before the policy change. Third, the reform's mechanism—introducing a standardized credit score for Standard loans only—is specific to the treated group and orthogonal to any structural feature of Express lending.

2 Hypothesis Development

Specialization Drivers

The first set of hypotheses concerns the extensive margin of specialization: how lenders enter and exit industries. The first driver of specialization is private information. When information is costly to acquire, lenders face a trade-off between diversification, which can reduce failure risk and smooth returns (Diamond, 1984), and concentration, which can dominate when information rents are high or monitoring incentives are weak (Winton, 1999). Private information acquired through repeated lending within an industry can create an ex post information monopoly (Blickle et al., 2023; Paravisini et al., 2023; Petersen and Rajan, 1994), pushing lenders to specialize where their expertise raises screening and monitoring effectiveness and softens competition.

The countervailing force is standardized information, such as credit scores, collat-

eral valuations, and financial statements (Frame et al., 2001; Frame and Woosley, 2004; Hauswald and Marquez, 2003; Keys et al., 2012). Standardized signals are portable and broadly comparable across lenders. By compressing information asymmetries within an industry (Hauswald and Marquez, 2003), standardized information lowers the fixed cost of entering a new sector, raises contestability, and erodes the information rents that sustain concentrated lending, encouraging lenders to diversify.

Its effect on specialization hinges on whether the public signal substitutes for or complements private expertise. If the signal is sufficiently informative to stand in for much of the screening and monitoring advantage from specialization, the returns to industry-specific knowledge fall and lenders diversify (Hypothesis $H_{1.a}$). If, instead, the signal is an imperfect baseline that is most valuable when interpreted through experience, lenders can combine it with existing expertise to sharpen selection and monitoring, reinforcing comparative advantage and deepening specialization in industries where they are already strong (Hypothesis $H_{1.b}$).

Hypothesis $H_{1.a}$**.** *Dissemination of standardized information reduces specialization by compressing private information advantages and lowering entry costs across industries.*

Hypothesis $H_{1.b}$**.** *Dissemination of standardized information is used strategically to complement private expertise and increase specialization where information advantages are strongest.*

Information Substitution and Loan Performance

The substitution and complementarity predictions concern the extensive margin—whether lenders diversify or deepen their portfolios. But if standardized information substitutes for private expertise (Hypothesis $H_{1.a}$), the displacement also has consequences for loan outcomes. Where lenders had deep pre-policy industry knowledge, public signals crowd out the private screening advantages that sustained superior lending performance. The welfare cost concentrates in previously specialized industries because the marginal screening value of private information was highest there. Meanwhile, competitive entry by generalist lenders, now equipped with public scores, can compress pricing even as screening quality deteriorates—producing a wedge between the default and spread channels.

Hypothesis H_2**.** *If standardized information substitutes for private expertise, loan performance deteriorates in industries where lenders were previously most specialized, as public signals displace the screening advantages that sustained lending quality.*

Intensive-Margin Effects

Loan Performance

The remaining hypotheses concern the intensive margin: how contemporaneous specialization affects loan-level outcomes regardless of the information regime. While Hypothesis H_2 predicts that displacing private expertise harms outcomes in previously specialized industries, the following hypotheses test whether specialization itself—whatever its level—retains value as an information-acquisition channel. Industry-specific expertise improves the lender's ability to produce and interpret information. In screening, experience helps translate sector-relevant features, such as revenue seasonality and typical paths into distress, into a more precise assessment of borrower quality than a generalist relying mainly on financial statements and credit scores. In monitoring, the same expertise lowers the cost and raises the effectiveness of oversight by focusing attention on the operating metrics and covenant signals that matter in that sector, enabling earlier intervention when problems emerge and reducing expected losses (Blickle et al., 2025).

Hypothesis H_3**.** *Specialization provides industry-specific information that improves ex ante screening and ex post monitoring, thereby enhancing loan performance.*

Relationship Lending

Specialization also shapes relationship lending. Industry expertise and borrower-specific knowledge operate at different levels but are informational complements. Industry expertise helps a lender interpret firm-level signals, for example, whether a revenue decline is idiosyncratic or reflects an industry-wide shock. Relationship lending, in turn, generates soft information over repeated interactions that is easier to process and act on when the lender understands the sector's operating and risk environment. Because each type of information raises the marginal value of the other, their joint returns are higher than either alone, strengthening specialized lenders' incentives to build and maintain long-term borrower relationships in their focus industries.

Hypothesis H_4**.** *Specialization and relationship lending exhibit informational complementarities, leading lenders to closely monitor borrowers over the long term through relationships with firms in their specialized industries.*

Loan Pricing

Specialization can affect loan pricing through two opposing channels. First, industry expertise lowers screening and monitoring costs and reduces risk uncertainty, pushing spreads down when competition forces pass-through (Cabossioras and Tielens, 2024; Giometti et al., 2024). Second, specialization can generate proprietary soft information through relationship lending, raising borrower switching costs and creating hold-up power for the incumbent lender, which can raise spreads or worsen renegotiation terms (He et al., 2024; Rajan, 1992; Santos and Winton, 2008). Repeat borrowers in specialized industries provide the sharpest test, because the lender holds both industry expertise and borrower-specific knowledge: if the combined advantage lowers costs, spreads fall; if it creates hold-up power, spreads rise.

Hypothesis $H_{5.a}$**.** *Specialization creates an information advantage that increases hold-up incentives and enables the extraction of monopoly rents.*

Hypothesis $H_{5.b}$**.** *Specialization lowers screening and monitoring costs, and competitive pressure passes these savings through to borrowers in the form of lower spreads.*

3 Specialization Measure and Data

3.1 Measuring Specialization

Since the benefits of specialization derive from comparative rather than absolute advantage, a concentration index such as the Herfindahl–Hirschman Index (HHI) does not capture the relevant variation. Beyond this, the analysis requires a measure that is (i) comparable across lenders of different sizes and across industries, and (ii) handles zero-lending cells to track entry into new sectors. The normalized revealed comparative advantage (NRCA) index of Yu et al. (2009), building on Balassa (1965), satisfies both requirements. Throughout the paper, I denote this index $Spec$.

The measure starts from a diversification benchmark. If lender j is fully diversified, its portfolio share in industry s equals the market-wide share of industry s:

$$\frac{L_{js}}{L_j} = \frac{L_s}{L} \tag{1}$$

where L_{js} is lender j's credit to industry s, $L_j \equiv \sum_{k \in S} L_{jk}$ is lender j's total credit, $L_s \equiv \sum_{b \in B} L_{bs}$ is total credit to industry s, and $L \equiv \sum_{b \in B} \sum_{k \in S} L_{bk}$ is total market

credit. Comparing lender j's actual allocation to the diversification benchmark and normalizing by total market credit L yields the specialization index:

$$Spec_{js} = \frac{L_{js}}{L} - \frac{L_s}{L} \cdot \frac{L_j}{L} \tag{2}$$

Factoring out the lender's market share clarifies the underlying mechanics of the index:

$$Spec_{js} = \frac{L_j}{L}\left(\frac{L_{js}}{L_j} - \frac{L_s}{L}\right) \tag{3}$$

As this decomposition illustrates, the index is "scale-invariant" in the specific sense defined by Yu et al. (2009): it is invariant to uniform scaling of the overall market size, L. However, it is intentionally not neutral to individual lender size, L_j. Instead, it explicitly weights a lender's internal portfolio-share deviation ($\frac{L_{js}}{L_j} - \frac{L_s}{L}$) by the lender's overall market footprint ($\frac{L_j}{L}$). Two lenders with identical portfolio allocations but different total lending volumes will therefore yield proportionally different $Spec$ values: a large lender's industry tilt has greater market impact than the same tilt by a small lender, and the measure reflects this.

A positive $Spec_{js}$ indicates that lender j over-invests in industry s relative to the market benchmark, while a negative value indicates under-investment. The index takes values in $[-0.25, 0.25]$, with a symmetric distribution centered at zero.[7]

To illustrate, suppose total market lending is \$100 billion, lender j lends \$5 billion overall (5% of the market), and industry s totals \$10 billion (10% of the market). Under full diversification, lender j's expected allocation to industry s would be \$0.5 billion (0.5% of the market). If the lender instead allocates \$1 billion to that industry (1% of the market), then

$$Spec_{js} = 0.01 - 0.10 \times 0.05 = 0.005, \tag{4}$$

indicating \$0.5 billion more lending to industry s than diversification would predict.

[7] Let $s_{js} \equiv L_{js}/L$ and $s_s \equiv L_s/L$. Then

$$Spec_{js} = s_{js} - s_s\left(\frac{L_j}{L}\right)$$

If lender j lends only to industry s, then $L_j = L_{js}$ so $L_j/L = s_{js}$ and $Spec_{js} = s_{js} - s_s s_{js} = s_{js}(1 - s_s)$. Since $L_s \geq L_{js}$ implies $s_s \geq s_{js}$, for fixed s_{js} this is maximized at $s_s = s_{js}$ (industry s borrows only from lender j), giving $Spec_{js} = s_{js}(1 - s_{js}) \leq 1/4$ (attained at $s_{js} = 1/2$).

The choice of normalization is critical for the extensive-margin analysis, which requires comparing specialization across industries and across lenders of different sizes. A natural alternative measure is the simple difference between a lender's portfolio share and the market share:

$$RCA3_{js} = \frac{L_{js}}{L_j} - \frac{L_s}{L} \tag{5}$$

This measure has been widely applied in recent studies of lender specialization (Duquerroy et al., 2022; Giometti et al., 2024; Paravisini et al., 2023). It sums to zero across industries within a lender, so one can compare a given lender's specialization across its industries. However, this measure has a critical issue: it does not sum to zero (or any constant) across lenders within a given industry. The cross-lender sum depends on the distribution of lender sizes:

$$\sum_{j \in B} RCA3_{js} = \sum_{j \in B} \frac{L_{js}}{L_j} - |B| \cdot \frac{L_s}{L} \tag{6}$$

where $|B|$ is the number of lenders. The first term is a size-weighted combination of lender portfolio shares that varies across industries, making cross-industry comparisons unreliable. In particular, smaller lenders mechanically produce larger portfolio shares in any given industry, inflating measured specialization for small lenders relative to large ones. Because the extensive-margin analysis compares how specialization shifts across industries in response to the information shock (including entry by small lenders into new sectors), this size bias would directly contaminate the estimates.

The $Spec$ index in Eq. (2) eliminates this problem by dividing by total market credit rather than lender-level credit. The resulting measure sums to zero across both lenders and industries. Internet Appendix A proves these properties formally and shows that two commonly used alternatives, the standard Balassa index and the difference measure in Eq. (5), fail one or both comparability conditions. Figure 2 illustrates the distributions of each specialization measure. Only the normalized measure exhibits an approximately symmetric distribution, whereas the other measures are not comparable across industries.

The measure has an additional property relevant for the analysis: it captures zero-

investment decisions. If lender j does not originate loans in industry s, then

$$Spec_{js} = -\frac{L_s}{L} \cdot \frac{L_j}{L}, \tag{7}$$

recording the under-investment directly. This is critical for the extensive-margin analysis, where measuring entry into new sectors and exit from existing ones requires distinguishing zero lending from missing data.[8] One limitation of the signed index is that it sums to zero within a lender and therefore cannot serve as a lender-level summary of specialization intensity. For analyses of changes in specialization levels (Section 4), I therefore take the absolute value of the index, which captures both over- and under-investment relative to the benchmark.

3.2 Data and Sample Construction

The primary dataset comes from the SBA 7(a) loan program. The sample period runs from 2011:Q1, the quarter in which the Small Business Jobs Act increased SBA loan limits,[9] to 2019:Q4, just before the COVID-19 pandemic. The initial dataset comprises 3,005 lenders, including 2,525 FDIC-insured banks, 426 credit unions, and 54 non-bank financial institutions (SBA Loan Companies). I drop lenders with no originations during the sample period, observations that lack important lender or borrower information (no lender name or id), and loans with implausible terms (negative interest rates, zero principal, or zero maturity).

I exclude five rarely represented industries that together account for less than one percent of observations: Energy (North American Industry Classification System [NAICS] 21), Utilities (NAICS 22), Postal Services (NAICS 49), Management of Companies and Enterprises (NAICS 55), and Public Administration (NAICS 92).[10] I also exclude lenders that (i) lack Preferred Lender Program status,[11] (ii) issue only non-Preferred Lender Program credits, or (iii) lend to only one industry during the sample. The final sample includes 294,589 loan observations from 638 lenders.

Lenders report detailed loan-level information covering loan terms (annual per-

[8] Concentration measures such as the HHI cannot distinguish between zero investment and non-participation, since both yield the same index value.

[9] Effective October 8, 2010, the SBA permanently raised the 7(a) loan limit from $2 million to $5 million and the 504 loan limit to $5.5 million, with corresponding adjustments to maximum guaranty amounts.

[10] Their individual shares are 0.25%, 0.10%, 0.42%, 0.07%, and 0.02%, respectively.

[11] Preferred Lender Program status delegates most underwriting authority to the lender and provides faster processing, localized expertise, streamlined procedures, and flexible structuring.

centage rate, spread, principal, maturity), contract characteristics (guaranty share, rate type, credit type, collateral, secondary-market sale status), and ex post outcomes (charged-off status, paid-in-full status). The interest-rate spread is the difference between the loan's initial rate and the SBA base rate (three-month U.S. Treasury bill plus 300 basis points). Borrower attributes include geography (county and state), franchise status, legal form, and a firm identifier that links repeat borrowers (Internet Appendix B describes the construction of borrower identities and the relationship lending indicator). Borrowers' six-digit NAICS codes enable lenders to measure their relative exposure to specific industries. Following Blickle et al. (2023), I compute industry specialization at the two-digit level, which balances detail and statistical power while avoiding the sparsity that arises with finer classifications. All variable definitions are listed in Table 3.

Table 4 reports summary statistics by policy period (pre-policy versus post-policy) and loan type (control: Express versus treated: Standard Guaranty). Panel A shows that the two loan types differ in level: treated (Standard Guaranty) loans are larger ($179K versus $74K), carry longer maturities (127 versus 77 months), have higher guaranty shares (0.79 versus 0.50), and are almost entirely term loans, whereas Express loans are predominantly lines of credit (60–65%). Treated loans also have lower spreads (270 versus 355 basis points), consistent with the higher federal guarantee reducing lender risk. These patterns are stable across the pre- and post-policy periods. These level differences are absorbed by the lender-industry-subprogram fixed effects in the difference-in-differences design; identification requires parallel trends, not level equivalence (Section 1). Charge-off rates range from 5.5% to 8.1% across groups and periods, and the relationship-lending share ranges from 13.2% to 16.1%.

Panel B reports lender–industry specialization from the balanced panel. Two patterns foreshadow the main results. First, the average number of active industry pairs per lender-quarter increases for the treated group after the reform (1.73 to 1.84) but declines for the control group (2.30 to 2.03), consistent with treated lenders entering new industries. Second, within active portfolios, the absolute specialization index declines sharply for treated lenders (0.054 to 0.038) but is essentially flat for control lenders (0.037 to 0.035), indicating that treated lenders reallocated credit toward the market benchmark. Both patterns preview the extensive-margin results in Section 4.

For the loan-level intensive-margin analyses, I scale $Spec$ by 100 so that a one-unit increase corresponds to a 0.01 increase in the raw index.

4 Lending Specialization on the Extensive Margin

This section documents lender specialization in SBA lending and estimates how the 2014 credit-scoring reform shifted specialization on the extensive margin.

4.1 Descriptive Evidence on Specialization in SBA Lending

The five most common industries in the SBA program are Accommodation and Food Services (NAICS 72; 17%), Motor Vehicle and Parts Dealers (NAICS 44; 12%), Health Care and Social Assistance (NAICS 62; 10%), Other Services (NAICS 81; 9%), and Professional, Scientific, and Technical Services (NAICS 54; 7%). Figure 3 plots quarterly loan shares for these industries. The series is essentially flat, confirming that changes in lender-level specialization reflect supply-side portfolio choices rather than shifts in industry loan demand.

To illustrate lender-level specialization, I examine the two largest SBA lenders. Wells Fargo originated $13.4 billion in SBA loans during the sample period; even for a large, diversified institution, it exhibits systematic tilts toward Professional Services and Health Care and away from Accommodation and Food Services (Internet Appendix Figures IA.1–IA.2). After the 2014 reform, both lenders' $Spec$ scores move closer to zero, and Live Oak begins originating in industries where it previously had no presence (Internet Appendix Figures IA.3–IA.4), supporting the view that credit scores lower barriers to entry.

4.2 The Information Shock and Specialization

4.2.1 Specialization Shift

I begin by testing whether the information shock reduced specialization (Hypothesis H_1).

Before the estimation, I assess the parallel-trends assumption using the Rios-Avila et al. (2023) joint pre-trend test within the eight quarters preceding the policy (Internet Appendix Table IA.4). The test fails to reject for $|Spec|$ ($\chi^2(7) = 10.71$, $p = 0.152$), $Spec^2$ ($p = 0.457$), and HHI ($p = 0.838$). The design also maintains the no-anticipation assumption: although the SBA announced the mandate before its July 2014 implementation, credit scoring affects loan processing at the time of application rather than ex ante portfolio decisions, and the event-study pre-treatment coefficients show no evi-

dence of anticipatory adjustment. Figure 4 presents the event-study coefficients. It exhibits a pattern of declining industry specialization over time, reflecting reduced specialization incentives as lenders become more familiar with new industries through the credit-scoring policy.

Specifically, I estimate two-way fixed-effects regressions of the following form:

$$Y_{j,s,\ell,t} = \gamma_1 Treated_\ell \times Post_t + FE + \epsilon_{j,s,\ell,t} \tag{8}$$

Here, $Y_{j,s,\ell,t}$ is lender j's specialization measure in industry s within loan type ℓ at quarter t. As an alternative outcome, I use lender j's HHI for loan type ℓ at time t.[12] The sample is restricted to loans up to \$350,000, the threshold below which the SBSS mandate applies.

$Treated_\ell$ equals one for Standard Guaranty loans, which were subject to the credit-score screening starting in 2014:Q3. $Post_t$ is a binary indicator that equals one in periods after the policy intervention. The FE terms denote lender-industry-subprogram fixed effects and time fixed effects. $\epsilon_{j,s,\ell,t}$ is a disturbance term; standard errors are clustered at the lender level to account for within-lender correlation.

γ_1 measures the impact of the information shock on a lender's over- or under-investment in the treated loans. A negative γ_1 supports Hypothesis $H_{1.a}$, indicating that the dissemination of information reduces deviation from complete diversification. Conversely, a positive γ_1 would support Hypothesis $H_{1.b}$ that the additional hard information promotes further specialization across industries.

Table 5 presents the results from estimating Eq. (8) for lending-portfolio measures. The estimate in column 1 shows that lenders' cross-industry specialization declines, supporting Hypothesis $H_{1.a}$. Specifically, the average treatment effect on the treated suggests that the policy reduces specialization by 12.7 percent relative to its pre-treatment baseline.[13] This result is robust to the Callaway and Sant'Anna (2021) difference-in-differences estimator (Internet Appendix Table IA.3).[14] The estimate on

[12]Unlike the specialization measures, the HHI does not capture comparative advantage; it only reflects how concentrated a lender's portfolio is.

[13]This magnitude is calculated by dividing the treatment effect (-0.00193) by the pre-treatment (pre-2014:Q3) mean $|Spec|$ specifically for the treated group (Standard Guaranty loans $\leq \$350,000$), which is 0.0152: $0.00193/0.0152 = 0.127$. This pre-treatment baseline is naturally higher than the full-sample unconditional mean of 0.0104, because the full-sample summary statistics pool across the entire timeframe, including the post-treatment period where specialization significantly declined, as well as across the exempt Express loans.

[14]The CSDID robustness check clusters at the panel (lender-industry-subprogram) level, as required

the concentration measure (HHI) in column 3 is positive and marginally significant ($p < 0.10$), indicating that treated lenders' portfolios became more concentrated within the industries where they continued to lend.[15]

4.2.2 Decomposing the Shift in Specialization

What caused the reduction in specialization? As discussed in Section 3.1, the $Spec$ index reflects both a lender's internal portfolio deviation and its overall market footprint. Therefore, the overall reduction captures a decline in the aggregate volume of specialized lending. This decline is driven by three potential forces: mechanical changes in lenders' overall market shares, entry into new industries, and the reallocation of credit within active portfolios.

To understand the behavioral drivers, I focus on entry and reallocation. Both stem from the same mechanism: standardized credit scores compress the cost gap between familiar and unfamiliar industries, simultaneously lowering barriers to new-sector entry and reducing the information rents that sustained overweighted positions. Entry reduces specialization because initiating lending in a previously absent industry moves the index from its negative floor toward zero (Section 3.1). Reallocation reduces specialization because shifting credit away from overweighted industries toward the market benchmark lowers the index magnitude.

To quantify these two channels, I estimate the regression model in Eq. (8). To isolate the reallocation component, I estimate this model using the absolute specialization index as the dependent variable, restricted to the subsample of active lender-industry pairs where lending is positive ($L_{j,s,\ell,t} > 0$). By considering these active pairs, the coefficient γ_1 captures the average reduction in specialization within established relationships. To isolate the entry driver, I estimate a Linear Probability Model on the full sample, where the dependent variable is a binary indicator ($\mathbb{1}_{Active}$) equal to 1 if the lender has positive lending in the industry and 0 otherwise. In this specification, γ_1 measures the change in the probability of entering a new industry following the policy shock.

Table 6 reports the regression results. In column 1, the reallocation estimate is statistically significant and larger in absolute magnitude than the overall specialization shift

by the estimator's panel structure, whereas the TWFE specification clusters at the lender level.

[15]The HHI regression uses the lender–loan-type–quarter panel (43,406 observations) rather than the lender–industry–loan-type–quarter panel used for the specialization measures, because HHI is a lender-level concentration index that does not vary across industries.

in Table 5. This difference is partly mechanical: restricting to active lender-industry pairs concentrates the effect by excluding zero-lending cells. Column 2 shows that standardized credit scoring increased the probability of a treated lender entering a new industry by 3.43 percentage points.

Finally, I verify that these effects are not driven by changes in lenders' overall market shares by re-estimating the DID using the unweighted portfolio deviation ($RCA3_{js}$) from Eq. (5), which isolates internal portfolio-share deviation independently of lender scale. Within active portfolios, unweighted specialization fell by a significant 0.0262 (Table 7), confirming that lenders shifted their portfolio compositions toward diversification even conditional on size. The full-sample estimate is also negative (-0.0027, $p < 0.05$) but smaller, reflecting the unweighted measure's sensitivity to noise from small lenders in zero-lending cells. These results confirm that the aggregate shift in Table 5 reflects behavioral reallocation, not mechanical scale changes.

4.2.3 Persistence of Specialization

If specialization reflects durable information advantages, it should be highly persistent; the 12.7 percent decline documented above is more notable if it represents a shift in a sticky equilibrium. To test this, I estimate a two-step System GMM dynamic panel on the full balanced panel and the active-pairs subsample (Table 8). The AR(1) specification on the full panel (column 1) yields a lagged coefficient of 0.975 with a post-2014 interaction of -0.055 ($p < 0.01$), implying effective post-policy persistence of 0.920. However, the Arellano–Bond AR(2) test rejects in the AR(1) specifications ($p < 0.001$), indicating second-order serial correlation in the differenced residuals and potentially invalid lag(2) instruments.

To address this, columns (2) and (4) estimate an AR(2) model—including two lags of specialization and their post-period interactions—using instruments from lag(3) onward. The full-panel specification (column 2) passes the AR(2) test ($p = 0.407$) and marginally passes the Hansen J test ($p = 0.050$). While individual lag coefficients are less precisely estimated due to the additional parameters and deeper instrument lags, the long-run persistence—the sum of the two lag coefficients—is near unity in the pre-policy period (0.976) and declines modestly post-2014 (to 0.944). The active-only AR(2) specification (column 4) produces similar long-run estimates (pre: 0.998, post: 0.863) and passes Hansen J ($p = 0.164$). These patterns confirm that specialization is strongly persistent yet attenuated after the public-information expansion, suggest-

ing that private information advantages are partially eroded by transparency. Overall, lenders' specialization choices are highly persistent, and the post-2014 shift modestly accelerates mean reversion, indicating that standardized information erodes the private advantages that sustain concentrated portfolios.

4.2.4 Hard Information Shock and Loan Outcomes

I now examine how the information shock affects loan-level outcomes across three margins: long-run default rates to capture screening and monitoring quality, relationship lending to proxy monitoring incentives, and interest-rate spreads to assess pricing. Because standardized credit scores differ from private industry expertise in scope and portability, their effects may vary across these channels. I estimate the following regression:

$$Y_{i,j,s,\ell,t(T)} = \gamma_1 Treated_\ell \times Post_t + \gamma_2 \overline{Spec}_{j,s,\ell} + X_{i,j}^\top \Gamma + FE + \epsilon_{i,j,s,\ell,t} \tag{9}$$

Here, $Y_{i,j,s,\ell,t(T)}$ is the outcome for loan i, originated by lender j in industry s under subprogram ℓ at time t, tracked through maturity T. The controls $X_{i,j}$ include loan terms and borrower characteristics as defined in Table 3. Relationship lending and repeated-borrower indicators are included in the default regressions to capture prior-relationship effects. The key addition is $\overline{Spec}_{j,s,\ell}$, the pre-policy median specialization, held fixed to avoid the bad-control problem that would arise from including contemporaneous specialization (Caetano et al., 2022). Fixed effects for lender-time, region(state)-time, and industry-time tighten the specification. The parameters of interest are γ_1, the information-shock effect, and γ_2, the specialization effect on loan outcomes.

Before examining the results, I assess the parallel-trends assumption for loan outcomes using a TWFE event-study specification that replaces the single $Treated \times Post$ interaction with quarter-by-treatment indicators within an eight-quarter window around the policy (Internet Appendix Figure IA.6). A joint F-test of the seven pre-treatment interaction coefficients ($t = -8$ to $t = -2$, with $t = -1$ as reference) yields $F(7, 9418) = 2.17$ ($p = 0.033$) for charge-off rates and $F(7, 9418) = 4.32$ ($p < 0.001$) for spreads. For charge-off rates, no individual pre-treatment coefficient is statistically significant, and the marginal joint rejection ($p = 0.033$) reflects the test's high power with 292,794 observations rather than a systematic pre-trend. For spreads, the rejection

is driven by two isolated coefficients with opposite signs ($t = -5$: -0.12, $p = 0.028$; $t = -2$: $+0.12$, $p = 0.022$), indicating oscillation rather than a monotonic pre-trend. To assess robustness to pre-trend violations, I apply the Roth (2022) honest confidence intervals under relative-magnitudes restrictions (Internet Appendix Table IA.6). For charge-off rates, the honest CIs remain centered near zero at $\bar{M} \in \{1, 2\}$, confirming robustness. For spreads, the honest CIs include zero at $\bar{M} = 1$ ($[-0.272, 0.034]$) and widen at $\bar{M} = 2$ ($[-0.376, 0.123]$), so the spread DID should be interpreted with caution.[16]

Panel A of Table 9 reports the DID regression results of Eq. (9). In column 1, the policy reduces the default rate by approximately 1.0 percentage point ($\hat{\gamma}_1 = -0.010$, $p < 0.01$), reflecting standardized scores providing a baseline risk assessment that improves screening precision. In column 2, the estimate on relationship lending is positive but not statistically significant, indicating no detectable shift in relationship-lending behavior on average. The pre-policy specialization control ($\overline{Spec}$) is strongly positive and significant ($\hat{\gamma}_2 = 0.060$, $p < 0.01$), confirming that specialization and relationship lending are informational complements: lenders maintain longer-term borrower relationships in industries where they possess deeper expertise. Column 3 shows that the policy reduces interest-rate spreads by approximately 21.9 basis points ($p < 0.01$), though this estimate is sensitive to pre-trend assumptions: honest confidence intervals under moderate departures from exact parallel trends include zero (Internet Appendix Table IA.6), so the spread DID should be interpreted with caution.[17]

4.2.5 Information Substitution

To assess whether public information substitutes for or complements lender specialization, I augment Eq. (9) with a triple interaction, $\delta_1 \, Treated_\ell \times Post_t \times \overline{Spec}_{j,s,\ell}$, where δ_1

[16] With 292,794 observations and 9,419 clusters, the event-study F-test has substantial power to detect economically insignificant deviations from exact parallel trends. No individual pre-treatment coefficient exceeds 2 percentage points in magnitude for charge-off rates. For spreads, the two significant pre-treatment coefficients have opposite signs and magnitudes roughly symmetric around zero, consistent with transitory noise rather than a confounding trend. Internet Appendix Table IA.5 reports placebo timing tests that further support the parallel-trends assumption.

[17] The three outcome dimensions—default, relationship lending, and spreads—represent conceptually distinct economic channels (performance, monitoring, pricing). Their consistency in sign and significance across specifications mitigates multiple-comparisons concerns; moreover, with three simultaneous tests the Bonferroni-adjusted significance threshold is $0.05/3 \approx 0.017$; the default ($p < 0.01$) and spread ($p < 0.01$) estimates remain significant under this correction, while relationship lending is not significant even without adjustment.

captures the differential policy effect by pre-policy specialization level. Panel B of Table 9 presents these triple-interaction estimates. The coefficient on the triple interaction for default ($\hat{\delta}_1 = 0.037$, $p < 0.05$) indicates that a one-standard-deviation increase in pre-policy specialization raises post-reform defaults by approximately 0.7 percentage points (0.037×0.194), or 11 percent of the pre-policy default rate, partially offsetting the baseline policy gains. This finding supports Hypothesis H_2: in industries where lenders were most specialized before 2014, the introduction of standardized screening erodes private screening advantages and leads to worse ex post loan performance.

The information substitution operates asymmetrically across pricing and performance margins. While defaults rise in previously specialized industries, interest-rate spreads do not increase—the point estimate is negative but not statistically significant (-0.163, $p = 0.57$). This asymmetry reflects a competition channel: the same policy that erodes screening quality also lowers entry barriers into previously specialized markets. Generalist lenders, newly equipped with credit-score information, can contest these markets and bid down loan pricing. The net effect on pricing is a wash—the loss of information rents that would push spreads upward is offset by competitive entry that pushes them downward. In contrast, the default channel is not subject to this offset, because loan performance is determined ex post by screening quality rather than ex ante by market competition.

The result is information substitution: borrowers in previously specialized industries receive similar or lower interest rates but face higher default risk, as the loss of private information manifests in ex post performance rather than ex ante pricing. This pattern is robust to matching on borrower characteristics: on a CEM-matched sample within high-specialization industries, the triple interaction for spreads is large and negative (-0.895, $p < 0.01$), confirming that holding borrower composition constant, the pricing compression in specialized industries is even more pronounced (Table 17).

The estimate on relationship lending is negative (-0.024) but not statistically significant, providing directional but inconclusive evidence that the return to building relationship capital declines in previously specialized industries.

4.2.6 Monitoring and Screening

If public information substitutes for private expertise (Hypothesis H_2), the substitution should be strongest where private information is thinnest—among new borrowers without prior lending relationships. To test this, I augment Eq. (9) with an interaction

using a new-borrower indicator. Because repeat borrowers already embody information from prior interactions, this triple interaction isolates the incremental impact of hard information on screening at origination.

Panel C of Table 9 reports these results. The triple interaction for default is negative and significant ($\hat{\lambda}_1 = -0.018$, $p < 0.01$): the credit-scoring policy reduces long-run defaults by an additional 1.8 percentage points for new borrowers relative to repeat borrowers. The corresponding estimate for spreads is also negative and significant (-0.111, $p < 0.01$): new borrowers receive an additional 11.1 basis point reduction in interest rates. Economically, the credit-scoring policy improves ex ante risk assessment for applicants without prior lending ties, allowing lenders to screen more precisely and price more competitively for first-time borrowers. These patterns point to a screening channel in which public, standardized information substitutes for private, relationship-based knowledge at the point of origination. The benefits of public information are concentrated among borrowers for whom private information is least available, precisely where the screening value of credit scores is highest.

4.2.7 Matching Robustness

To address potential covariate imbalance between treated and control loans, I re-estimate the DID specifications on matched samples constructed via Coarsened Exact Matching (CEM) and Propensity Score Matching (PSM). CEM matches on log loan amount, maturity, rate type, repeated-borrower status, collateral, revolver status, corporate type, and franchise status, retaining 135,277 observations; PSM uses nearest-neighbor matching on the same covariates, yielding 92,960 observations. The guaranty ratio is excluded from matching because Express and Standard Guaranty loans have structurally different guarantee levels (50% versus 75–85%), which would eliminate common support. Table 17 reports the results. Panel A shows that the baseline DID estimates are broadly consistent across matching methods, although the default coefficient differs in sign between CEM ($+0.011$, $p < 0.05$) and PSM (-0.029, $p < 0.01$). This sign reversal reflects differences in sample composition: CEM retains a broader set of treated loans by coarsening matching cells, while PSM's nearest-neighbor algorithm selects a narrower subsample that tilts toward lower-risk borrowers. Because the DID estimate captures the *change* in default conditional on matched observables, a positive CEM coefficient suggests the policy admits marginally riskier borrowers into the treated pool, whereas the negative PSM coefficient reflects improved screening among

closely matched pairs. The spread reduction is robust and significant in both samples (-0.286 CEM, -0.107 PSM, both $p < 0.05$). Panel B reports the triple interaction with pre-policy specialization. On both matched samples, the triple interaction for spreads is large and negative (-0.895 CEM, -0.712 PSM, both $p < 0.01$), confirming that the pricing compression in previously specialized industries is robust to controlling for observable borrower characteristics.

5 Specialization Effect on the Intensive Margin

The previous section established that public information erodes specialization on the extensive margin. This section asks whether the remaining specialization still creates value on the intensive margin. Unlike the extensive-margin analysis, which holds specialization at its pre-policy median ($\overline{Spec}$) to avoid the bad-control problem, the intensive-margin analysis uses contemporaneous specialization ($Spec_{j,s,\ell,t}$) as the regressor of interest, with an IV addressing the resulting endogeneity.

5.1 Specialization on Loan Outcomes

The intensive-margin hypotheses predict that specialization improves loan performance (Hypothesis H_3), strengthens relationship lending (Hypothesis H_4), and reduces spreads through lower screening and monitoring costs (Hypothesis H_5). To test these predictions, I estimate:

$$Y_{i,j,s,\ell,t(T)} = \beta_1 Spec_{j,s,\ell,t} + X_{i,j}^{\top}\Gamma + FE + \epsilon_{i,j,s,\ell,t} \tag{10}$$

The outcomes $Y_{i,j,s,\ell,t(T)}$ are the same as in Eq. (9): loan default (tracked through maturity T), relationship lending, and interest-rate spread.[18] The key difference is that $Spec_{j,s,\ell,t}$ is the contemporaneous specialization index (2), scaled $\times$ 100, rather than the pre-policy median. Controls $X_{i,j}$ are as before (Table 3). Fixed effects include subprogram-time, lender-time, industry-time, and state-time.

Table 10 reports the OLS results. In column 1, a one-unit increase in $Spec$ (corresponding to a 0.01 increase in the raw index) is associated with a 1.07 percentage point

[18]The argument T is relevant only when the outcome is loan default, which is observed over the life of the loan through maturity T.

lower default rate ($\hat{\beta}_1 = -0.0107$, $p < 0.01$), or roughly 19.5 percent of the mean default rate,[19] supporting Hypothesis H_3. Column 2 shows that specialized loans are 4.55 percentage points more likely to involve relationship lending ($p < 0.01$), supporting Hypothesis H_4: industry expertise strengthens monitoring and facilitates long-term borrower relationships. Column 3 indicates that spreads are 12.5 basis points lower in specialized industries ($p < 0.05$), supporting Hypothesis $H_{5.b}$: specialization lowers screening costs, and competitive pressure passes the savings through to borrowers.

5.1.1 Instrumental Variable Estimation

The OLS estimates may be biased in either direction. If lenders specialize in industries where unobserved risk rises or borrower composition deteriorates, OLS understates performance gains; if lenders concentrate where unobserved quality is better, OLS overstates them. The error term decomposes as

$$Y_{i,j,s,\ell,t} = \beta_1 Spec_{j,s,\ell,t} + X_{i,j}^{\top}\Gamma + FE + \underbrace{\phi_{j,s,\ell,t} + \rho_{i,j,s,\ell,t}}_{\epsilon_{i,j,s,\ell,t}}, \quad \text{Cov}[Spec_{j,s,\ell,t}, \phi_{j,s,\ell,t}] \neq 0 \quad (11)$$

where $\phi_{j,s,\ell,t}$ is a lender-industry unobservable confounded with industry shocks and lender-specific knowledge, and $\rho_{i,j,s,\ell,t}$ captures loan-level unobservables assumed uncorrelated with industry-wide specialization.

To isolate exogenous variation, I use a Bartik shift-share instrumental variable (Bartik, 1991). The instrument interacts each lender's pre-policy (pre-2014) industry portfolio weights with the leave-one-out national industry-quarter share of SBA lending, which captures aggregate reallocations across industries driven by macroeconomic demand rather than individual lender decisions. By leaving the focal lender out, the instrument avoids the mechanical link between the lender's own shocks and the national shift (Borusyak et al., 2025; Goldsmith-Pinkham et al., 2020). The identifying variation comes from common national lending reallocations interacted with predetermined exposure patterns; the excluded variation consists of idiosyncratic lender-industry movements (borrower composition, local conditions, screening intensity) that are likely correlated with outcomes. This strategy is conceptually distinct from the DID, which exploits treated-versus-control subprogram comparisons around 2014. Full details are in

[19] In the SBA 7(a) program, the mean default rate is 5.5 percent, ranging annually from 2.7 percent to 6.1 percent.

Internet Appendix C.

The first-stage results (Table 11) show that the instrument strongly predicts specialization, with Kleibergen–Paap F-statistics of 38–39, well above the Stock–Yogo critical value of 16.38. Exogeneity is a maintained assumption justified by the predetermined exposure shares and leave-one-out construction, not by the F-statistics.

The IV coefficient is interpretable as the causal effect of a marginal increase in specialization induced by national industry-quarter reallocations, for lenders whose portfolios respond to these shifts given their pre-policy footprint. The estimates speak to Hypotheses H_3–H_5: if an externally driven increase in specialization leads lenders to deploy more industry-specific knowledge, the IV should show lower defaults, more relationship lending, and lower spreads.

Panel A of Table 12 reports the IV estimates. Specialization functions as an information technology that creates value on three margins. First, specialized loans exhibit 1.23 percentage point lower default rates (OLS: 1.07 percentage point), primarily identified from capital-intensive sectors where expertise in asset valuation aids screening (as the Rotemberg decomposition below confirms). Second, relationship lending is more prevalent in specialized industries (6.50 percentage points; OLS: 4.55 percentage points), reflecting higher returns to monitoring and information reuse. Third, spreads are -11.6 basis points lower for specialized loans, though not statistically significant at conventional levels (OLS: -12.5 bps, $p < 0.05$), directionally aligned with industry-specific signals shrinking the uncertainty premium. The IV effects corroborate the OLS estimates: after purging selection, the IV reveals lower defaults and greater relationship lending, reinforcing the specialization-as-information-acquisition view. The IV spread estimate is attenuated and imprecise relative to the OLS, likely reflecting the limited pricing variation available under SBA interest rate caps.

Panel B of Table 12 employs the IV with a new-borrower interaction to distinguish between screening and monitoring. The baseline IV estimate indicates that specialization lowers default (-0.018, $p < 0.01$) and reduces spreads (-0.224, $p < 0.01$). The spread result for repeat borrowers directly tests the hold-up versus informational complementarity hypotheses (Hypotheses $H_{5.a}$–$H_{5.b}$): because repeat borrowers in specialized industries face a lender with both industry expertise and borrower-specific knowledge, the sign of the spread effect discriminates between the two channels. The negative coefficient supports Hypothesis $H_{5.b}$—industry expertise and borrower-specific knowledge are informational complements that reduce spreads—and rejects Hypothesis $H_{5.a}$, which predicts higher spreads from hold-up power. The new-borrower inter-

action for spreads is positive and marginally significant ($+0.132$, $p < 0.10$): the spread reduction is smaller for new borrowers ($-0.224 + 0.132 = -0.092$ in net), indicating that repeat borrowers receive an additional pricing benefit from relationship-specific information. The new-borrower interaction for default is near zero and not significant, suggesting that the screening channel operates primarily through pricing rather than ex post performance.

The Kleibergen–Paap F-statistic for Panel B is 13.59, which exceeds the Stock–Yogo 10% critical value of 7.03 for two endogenous regressors but falls below the single-endogenous threshold of 16.38. To verify that the results are not driven by weak identification, I compute Anderson–Rubin (AR) statistics, which are robust to arbitrarily weak instruments. The AR test rejects the joint null that both coefficients equal zero for the charge-off equation ($\chi^2(2) = 7.21$, $p = 0.027$), confirming that the IV results hold under weak-instrument-robust inference. For spreads, the AR test does not reject ($\chi^2(2) = 3.70$, $p = 0.157$). This weaker result for spreads is expected on institutional grounds: SBA interest rate caps constrain lenders' pricing discretion, compressing the range of observed spreads and limiting the variation available for identification. Because the maximum allowable rate is set as a fixed markup over the prime rate and does not vary with industry-level specialization, the Bartik instrument—which exploits differential exposure to national industry shifts—has limited ability to generate spread variation beyond what the rate ceiling permits. Consequently, the IV is better suited to detect effects on outcomes with greater scope for lender discretion, such as default rates and relationship-lending decisions, than on administratively constrained pricing margins.

To assess the sensitivity of the IV estimates to industry composition, I compute Rotemberg weights following Goldsmith-Pinkham et al. (2020). Table 13 reports the decomposition. All 19 industry weights are positive (no negative weights), ensuring that the IV estimate is a proper weighted average of industry-specific causal effects. However, Transportation (NAICS 48) accounts for 65.6 percent of the total weight, with the top five industries comprising 95.2 percent. To evaluate whether this concentration drives the results, I re-estimate the IV dropping each industry one at a time (Table 14). Excluding Transportation attenuates the charge-off coefficient substantially (from -0.012 to -0.004) and the estimate loses significance, while dropping any other single industry has negligible impact. This indicates that the IV is primarily identified from cross-lender variation in Transportation specialization.

Two observations mitigate this concentration concern. First, Transportation's just-

identified estimate ($\hat{\beta}_{\text{default}} = -0.018$) is directionally aligned with the aggregate IV (-0.012) and with Construction (-0.032), the second-largest contributor; the dominant industry does not produce an anomalous effect. Second, the instrument remains strong without Transportation (Kleibergen–Paap $F = 172.7$), and the 95 percent confidence interval excluding Transportation ($[-0.031, +0.023]$) includes the baseline estimate, indicating that the attenuation reflects reduced precision rather than a conflicting effect. Transportation's large weight arises because it experienced the largest national-level reallocation in SBA lending during the sample period; under the exposure-based framework of Borusyak et al. (2025), this makes it the most informative industry for the shift-share design.

5.1.2 Alternative IV: Express-Derived Shifts

The weight concentration carries economic content that sharpens the paper's information-based interpretation. To test whether the results are an artifact of Transportation's dominance, I construct an alternative shift-share instrument using industry shifts derived exclusively from Express lending, which was exempt from the 2014 credit-scoring reform and therefore free of circularity between the reform and the shift component (Internet Appendix D provides the formal construction and exclusion restriction).

Internet Appendix Table IA.1 compares four specifications for the charge-off outcome. The baseline IV and Express-only IV agree on sign: specialization reduces defaults (-0.012 pooled IV, -0.022 Express IV on the Express sample). The Express-shift IV applied to the Standard sample reverses sign ($+0.056$, $p = 0.027$), though this estimate is not robust to weak-instrument inference (Anderson–Rubin $p = 0.070$, 95% CI includes zero; first-stage Kleibergen–Paap $F = 13.2$). The reversal is explained by a complete redistribution of Rotemberg weights. Under pooled shifts, Transportation accounts for 65.6 percent of the identifying weight; under Express shifts, Transportation drops to 1.1 percent, while Accommodation (35.5 percent) and Other Services (25.4 percent) dominate (Internet Appendix Table IA.2). The two instruments identify specialization effects from fundamentally different industries, and the divergence maps onto a clear distinction in how private information operates across sectors.

In capital-intensive sectors with standardized collateral (Transportation, Construction), specialization provides expertise in asset valuation: knowing how to appraise fleet values, assess equipment depreciation, and evaluate residual worth allows specialized lenders to screen more precisely and recover more in default. This expertise is

difficult to replicate with a credit score alone, so the information advantage translates directly into lower defaults ($\hat{\beta}_{\text{default}} = -0.018$ for Transportation, -0.032 for Construction). In labor-intensive sectors with idiosyncratic, demand-driven risk (Accommodation, Other Services), specialization generates a different trade-off. A lender concentrated in restaurants accumulates expertise in that sector's operating environment, but also concentrates exposure to correlated shocks: when local demand weakens, simultaneous stress across the portfolio can overwhelm the screening advantage. The Express-shift IV, which draws primarily from these sectors, yields a positive point estimate for defaults ($\hat{\beta} = +0.056$), though this estimate is not robust to weak-instrument inference (Anderson–Rubin $p = 0.070$, 95% CI includes zero), suggesting insufficient power to distinguish the effect from noise rather than a definitive contradictory result.

The net effect of specialization on defaults thus depends on the balance between two forces: an information advantage from better screening through industry expertise, and a concentration cost from correlated portfolio exposure to sector-wide shocks. Where collateral is standardized and verifiable, the information channel dominates. Where risk is demand-driven and collateral is thin, concentration exposure can offset or reverse the screening gains.

Two results anchor this interpretation. First, the relationship-lending channel—the most direct test of specialization's information-acquisition value—is confirmed under both IV specifications ($+0.065$, AR confidence interval excludes zero), establishing that specialization generates monitoring value regardless of the industry's risk structure. Second, the OLS estimates, which weight all industries proportionally without depending on any single industry's identifying variation, show consistent negative effects on defaults (-0.011, $p < 0.01$), indicating that on average the information advantage outweighs concentration costs. The IV and Express-shift analyses refine this average by revealing the industry conditions under which the balance tilts.

A potential concern with charged-off outcome regressions is right-censoring: loans originated later in the sample period may not have had sufficient time to resolve, mechanically depressing observed default rates. In the estimation sample, 8.98 percent of loans carry an "EXEMPT" (unresolved) status, but this share rises to 24.31 percent among post-period treated loans compared with 8.18 percent among post-period control loans, creating differential censoring across the treatment margin. To address this, I re-estimate the OLS and IV charge-off specifications restricting the sample to resolved loans only—those with a final status of charged-off, paid-in-full, or cancelled (Table 15). Dropping unresolved loans removes 9 percent of observations but strengthens

the results: the OLS specialization coefficient is -0.012 ($p < 0.01$), matching the baseline, and the IV estimate strengthens from -0.012 to -0.015 ($p < 0.01$). These results confirm that right-censoring, if anything, attenuates the baseline estimates, and the specialization–default relationship is robust to restricting the sample to fully resolved loans.

As a complementary approach, I impose a seasoning restriction, retaining only loans originated before 2017:Q1 (the midpoint between the 2014:Q3 policy date and the 2019:Q4 sample end), so that every loan has at least three years to reach a terminal status. This restriction reduces the unresolved share from 24.31 to 13.68 percent among post-period treated loans and from 8.18 to 2.49 percent among controls, substantially narrowing the differential censoring gap. Table 16 reports the results. The OLS charge-off coefficient on the seasoned sample (-0.007, $p = 0.016$; $N = 196{,}312$) is smaller in magnitude than the baseline (-0.011) but remains negative and significant, confirming that specialization lowers defaults even after removing potentially censored late-sample loans. The IV estimate is directionally consistent (-0.006) though imprecise ($p = 0.21$), with a strong first stage (Kleibergen–Paap $F = 48.0$). The attenuation relative to the baseline reflects the mechanical loss of post-period variation when later cohorts are excluded, not a sign reversal. The repeated-borrower and spread coefficients retain their baseline signs and significance ($+0.035$, $p < 0.01$; -0.139, $p < 0.01$).

6 Conclusion

This paper asks why lenders concentrate portfolios in specific industries and what happens when a new source of public information compresses the cost advantage of industry expertise. The SBA's 2014 credit-scoring mandate provides a natural experiment to answer both questions.

On the extensive margin, the information shock reduces industry specialization by 12.7 percent relative to its pre-reform average, driven by entry into new sectors and reallocation away from overweighted industries. Both channels reflect the same mechanism: standardized credit scores compress the cost gap between familiar and unfamiliar industries, simultaneously lowering barriers to entry and eroding the information rents that sustained concentrated portfolios.

The reform's benefits are unevenly distributed. A one-standard-deviation increase in prior specialization raises post-reform defaults by 11 percent of the pre-policy de-

fault rate, as public information crowds out the private screening expertise that previously contained risk. Spreads do not widen in these industries, as competitive entry by generalist lenders offsets the loss of information rents, though the spread DID is sensitive to pre-trend assumptions and should be interpreted with caution. The asymmetry—higher default risk without compensating spread increases—illustrates an adverse consequence of public information: borrowers in previously specialized industries lose the protection of expert screening while gaining no pricing benefit.

On the intensive margin, specialization still creates value where it persists. Instrumented with a Bartik shift-share IV, a one-unit increase in specialization lowers defaults by 1.2 percentage points—primarily in capital-intensive sectors with verifiable collateral—and raises relationship lending by 6.5 percentage points. Repeat borrowers in specialized industries pay 22.4 basis points less—though this estimate is sensitive to weak-instrument inference—reflecting informational complementarities rather than hold-up (Rajan, 1992).

These gains are not uniform across industries. Specialization generates both an information advantage—better screening through industry expertise—and a concentration cost—correlated portfolio exposure to sector-wide shocks. In capital-intensive sectors with verifiable collateral, expertise in asset valuation dominates and defaults fall. In labor-intensive sectors with demand-driven, correlated risk, concentration exposure can offset the screening advantage. The relationship-lending channel is confirmed under both IV specifications, establishing that specialization generates monitoring value regardless of the industry risk environment; it is the default channel that is industry-dependent.

The central insight is that public information substitutes for, but does not eliminate, private industry expertise. Standardized signals expand access and reduce screening costs, yet the resulting diversification erodes specialized knowledge unevenly—most acutely where prior expertise was deepest. The productive role of specialization persists in industries where collateral is complex and monitoring and relationship lending complement hard information, but its value is attenuated in sectors where portfolio concentration amplifies correlated risk. Information policy should therefore expand transparency while recognizing that the net benefit depends on whether the displaced private expertise served primarily a screening function—where public signals can substitute—or a monitoring and asset-valuation function, where it cannot.

Two questions remain open. First, does the decline in specialization stabilize or deepen as lenders accumulate experience with public scores? Second, how does the

shift from private to public information affect the distribution of credit access and default risk across borrower segments beyond the SBA market?

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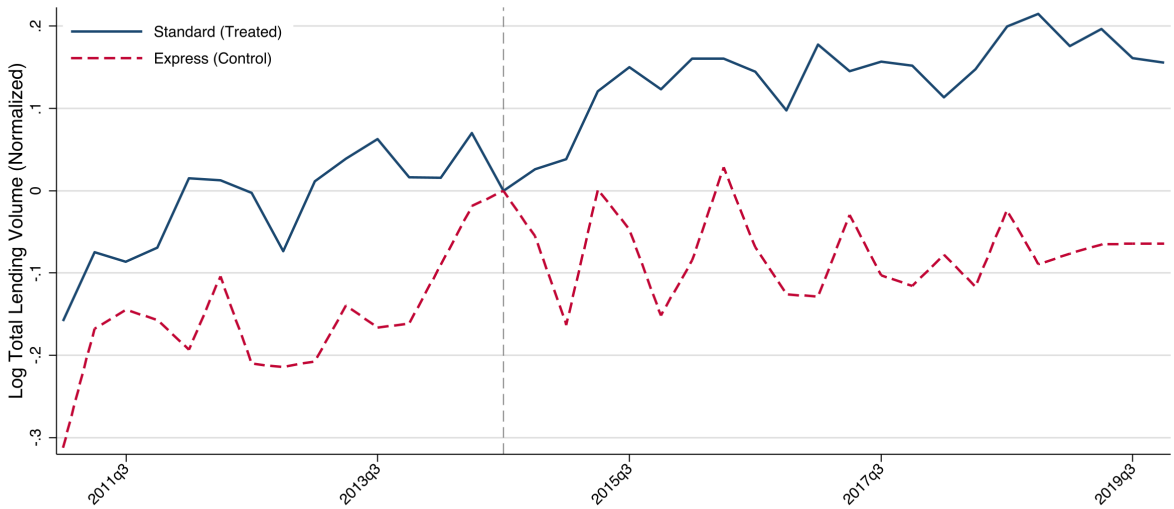


Figure 1: Parallel Trends: Lending Volume

Notes: This figure plots the mean log lending volume, ln (loan amount + 1), by calendar quarter separately for Standard loans (treated) and Express loans (control). The data are first aggregated to the lender–quarter–loan-type level by summing loan amounts within each cell, then averaged across lenders within each quarter and loan type. Both series are normalized to zero at the reform quarter (2014 Q3), marked by the vertical line.

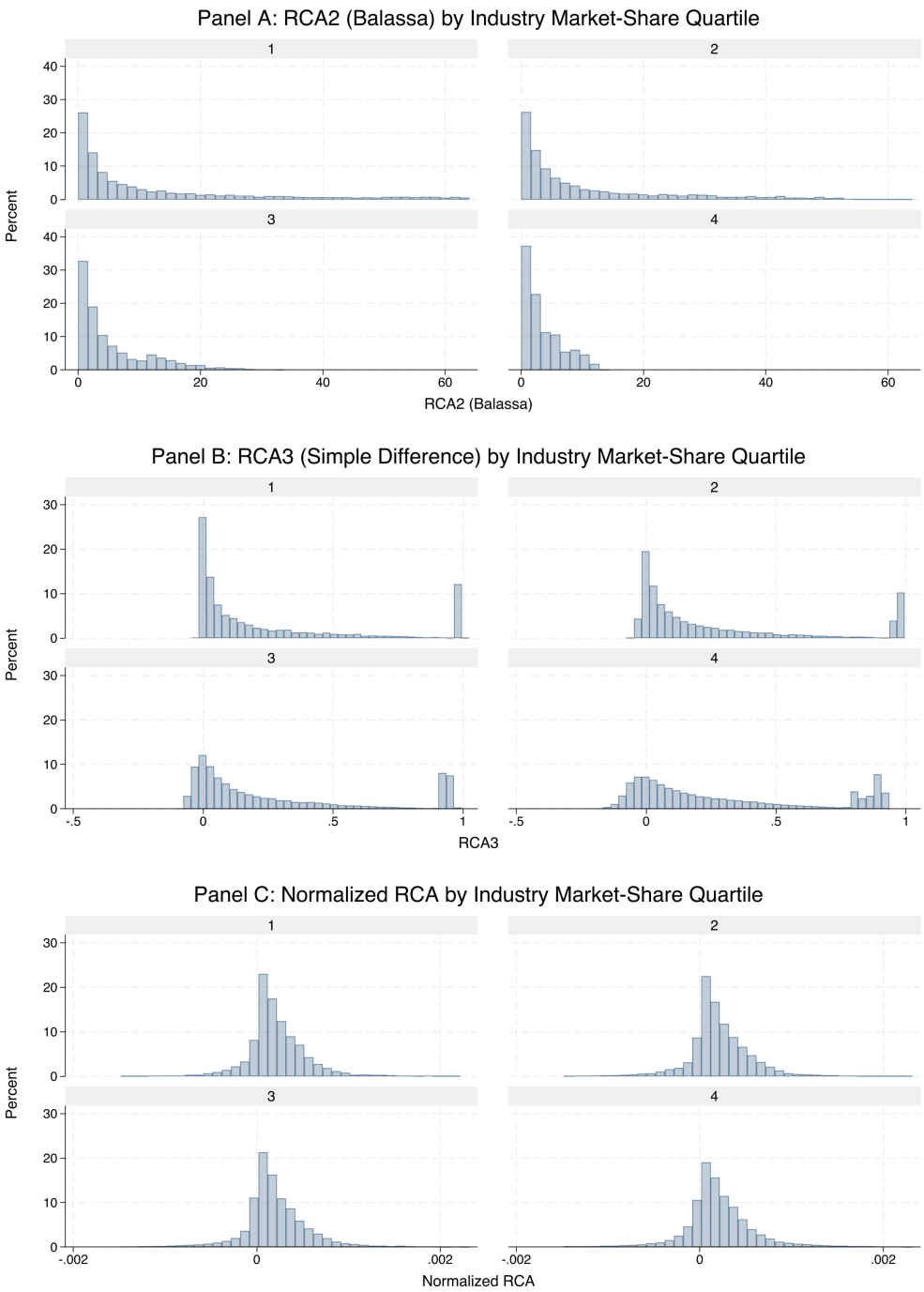


Figure 2: Distributions of lending specialization measures. The figure plots the empirical distributions of RCA2 (panel A; Balassa index), RCA3 (panel B; simple difference), and *Spec* (panel C; NRCA).

Notes: Panel A reports RCA2 (Balassa, 1965); panel B reports RCA3 (Blickle et al., 2023); and panel C reports *Spec* (NRCA; Yu et al., 2009). Each measure is computed at the lender-industry level. Industries are grouped into quartiles based on their market share, where 1 denotes the lowest-share quartile and 4 denotes the highest-share quartile.

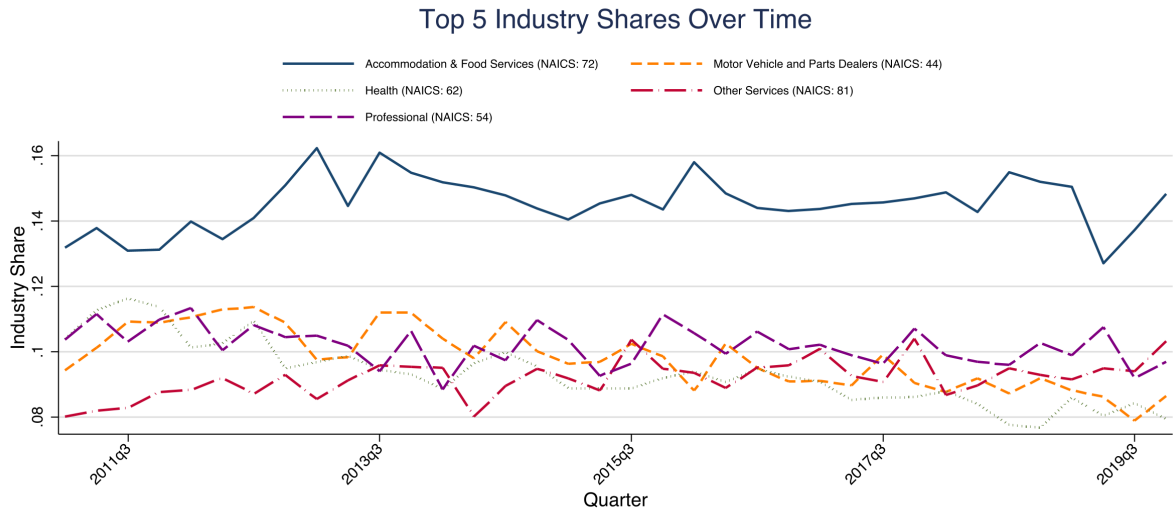


Figure 3: Top 5 Industry Loan Share

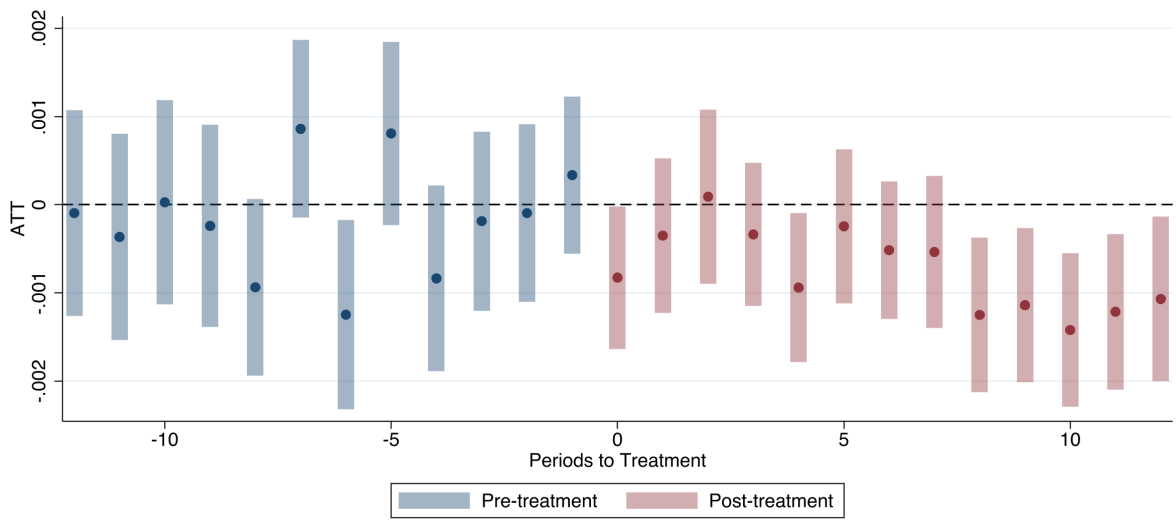


Figure 4: Information Shock on Specialization - Event Study

Notes: Event time measured in quarters relative to policy. Estimates shown with 95% confidence intervals. The CSDID pre-trend test of Rios-Avila et al. (2023), with standard errors clustered at the lender level, fails to reject that all seven estimable pre-treatment ATTs equal zero ($\chi^2(7) = 10.71$, $p = 0.152$); see Internet Appendix Table IA.4.

Table 1: SBA 7(a) Small and Express Program Description

Program	Credit Type	Loan Size	Guaranty Ratio	Maturity	Interest Rate Spread	Guarantee Fee
Standard Guaranty (Small)	Term Loan	\$350,000	85% ($\leq$\$150K) / 75% (>\$150K)	7–25 years	2.25%–4.75%	0.25%–3.75%[1]
	Credit Line[2]	\$350,000	85% ($\leq$\$150K) / 75% (>\$150K)	10 years	2.25%–4.75%	
SBA Express	Term Loan	\$350,000	50%	7–25 years	4.5%–6.5%	
	Credit Line	\$350,000	50%	7 years	4.5%–6.5%	

[1] Fee varies by loan size and maturity: 2% of guaranteed portion for loans $\leq$\$150,000 (lender retains 25%); 3% for \$150,001–\$700,000 (maturity >12 months); 3.5% on the first \$1M plus 3.75% above \$1M for \$700,001–\$5M; 0.25% for maturity $\leq$12 months.
[2] Credit Lines in 7(a) Small and Regular Programs are also called CAPLines.

Table 2: SBA 7(a) Credit-Scoring–Based Screening: Before vs. After 2014

Dimension	**Before 2014**	**After 2014 (Credit-Score Screening)**
Policy status	No SBA-mandated score screen for 7(a) Small Loans; lenders used standard underwriting (SOP-driven) and their own scoring as desired.	SBA requires a credit-score prescreen (SBSS) for non-Express 7(a) loans $\leq$ \$350,000 ("7(a) Small Loans"). If the score is acceptable, lenders may use a simplified credit analysis.[a,b]
Scope (coverage)	All 7(a) loans under standard / CLP / PLP procedures; no SBA credit-score requirement specific to small loans.	Applies to *non-Express* 7(a) loans $\leq$ \$350,000 submitted via E-Tran under "7(a) Small Loans."[a,b]
Screening criteria	No SBA scoring; full credit memo and standard documentation from lender.	SBA generates an SBSS score (from consumer & business bureau plus application data). Acceptable score $\Rightarrow$ shorter, simplified lender analysis; below cutoff $\Rightarrow$ submit as standard (large) 7(a) or use Express (if eligible).[b,c]
Minimum SBSS cutoff	Not applicable.	SBA sets and updates the minimum acceptable SBSS for 7(a) Small Loans (historically around 140 at rollout; updated over time).[a,d]

[a] CRS overview of the 7(a) program and expansion of credit-score screening to non-Express $\leq$ \$350k loans in 2014.
[b] OCC banker guidance (Dec. 2014) describing 7(a) Small Loans prescreen via SBA credit model, simplified analysis if score acceptable, and Express exclusion.
[c] SBA SOP 50 10 5(G): "SBA Credit Scoring is required on all 7(a) loans (non-Express) under \$350K as of July 1, 2014"; model had prior internal use.
[d] SBA SOP 50 10 7 indicates SBA posts and may adjust the minimum SBSS score over time.

Table 3: Variable Definitions

Variable	Definition
Panel A. Loan Characteristics	
Loan Amount	Total approved loan amount in USD
Spread	Interest rate spread above the SBA base rate: contractual rate minus (3-month T-bill + 300 bps), in percentage points
Maturity	Loan term in months
Guaranty Share	SBA guaranty amount divided by total loan amount
Collateral Backed	Indicator for loan secured by collateral
Line of Credit	Indicator for revolving credit line (vs. term loan)
Rate Type	Fixed or variable interest rate indicator
Franchisee	Indicator for franchise business
Corporation Type	Indicator for corporate borrower
Panel B. Loan Outcomes	
Default (Charge-off)	Indicator for loan default (charged off by lender)[a]
Relationship Lending	Indicator for same borrower–lender pair observed in a prior SBA loan (Internet Appendix B)[b]
New Borrower	Indicator for first-time borrower with lender (no prior SBA loan from same lender)[b]
Panel C. Specialization Measures	
$Spec$ (NRCA)	Normalized Revealed Comparative Advantage: lender j's deviation from expected lending in industry s, scaled by market total (Equation 2)
$\lvert Spec \rvert$	Absolute specialization; magnitude of deviation from proportional lending
$Spec^2$	Squared specialization
$\overline{Spec}$	Lender–industry pre-policy median of $Spec$
HHI	Herfindahl–Hirschman Index of lender portfolio concentration across industries
Panel D. Identifiers	
Industry	2-digit NAICS sector classification[c]
Lender ID	FDIC certificate number (banks), NCUA number (credit unions), or SBA-assigned name (non-depository lenders)

[a] Raw loan status: Paid in Full, Charged Off, Exempt (disbursed but unresolved), or Canceled. Default is coded as a binary indicator.
[b] Repeat borrowers identified via fuzzy string matching (Levenshtein distance) on business name and geography within lender, from program inception (1991). New Borrower = 1− Relationship Lending.
[c] SBA records contain 6-digit NAICS codes; the analysis aggregates to the 2-digit sector level.

Table 4: Summary Statistics by Treatment and Period

	Control				Treated			
	Pre		Post		Pre		Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Panel A-1: Loan Information								
Loan Amount (million USD)	0.074	(0.079)	0.068	(0.074)	0.179	(0.095)	0.182	(0.094)
Spread (%)	3.553	(1.645)	3.564	(1.835)	2.700	(0.715)	2.398	(0.916)
Maturity (months)	77.168	(33.900)	83.248	(32.202)	126.902	(73.263)	126.397	(63.311)
Guaranty Share	0.502	(0.031)	0.501	(0.016)	0.789	(0.057)	0.795	(0.058)
Collateral Backed	0.569	(0.495)	0.521	(0.500)	0.866	(0.340)	0.887	(0.317)
Line of Credit	0.605	(0.489)	0.649	(0.477)	0.000	(0.009)	0.000	(0.011)
Franchisee	0.027	(0.163)	0.032	(0.176)	0.124	(0.329)	0.199	(0.399)
Corporation Type	0.840	(0.367)	0.857	(0.350)	0.882	(0.322)	0.912	(0.283)
Panel A-2: Loan Outcome								
Default (Charge-off)	0.055	(0.227)	0.066	(0.247)	0.064	(0.245)	0.081	(0.273)
Relationship Lending	0.150	(0.357)	0.137	(0.344)	0.161	(0.368)	0.132	(0.338)
Observations	74,237		139,750		25,693		54,909	
Panel B: Lender–Industry								
Lender–Industry Pairs	7,039		7,039		6,503		6,503	
Lending Share	0.250	(0.273)	0.243	(0.272)	0.297	(0.286)	0.264	(0.268)
$\lvert Spec \rvert$ (incl. zeros)	0.010	(0.036)	0.009	(0.040)	0.014	(0.040)	0.010	(0.032)
$\lvert Spec \rvert$ (active only)	0.037	(0.073)	0.035	(0.087)	0.054	(0.085)	0.038	(0.065)
HHI	0.344	(0.335)	0.314	(0.337)	0.341	(0.346)	0.320	(0.329)
Active Pairs per Lender-Quarter	2.301	(3.533)	2.029	(3.488)	1.725	(2.760)	1.842	(3.094)
Observations	89,307		154,858		82,902		143,066	

Notes: Panel A entries report group means with standard deviations in parentheses. Groups are defined by post policy period (0=pre-2014, 1=post-2014) and control & treated (0=Express, 1=Standard Guaranty). Binary indicators are shown as shares. Loan amount is reported in millions of U.S. dollars (GrossApproval divided by 1,000,000). Panel B reports lender–industry level statistics from the balanced panel. Lender–Industry Pairs is the number of unique lender–industry combinations. Lending Share is the lender's within-industry share of its own total lending (active pairs only). $\lvert Spec \rvert$ is the absolute value of the specialization index defined in Eq. (2), reported for the full panel (including zero-lending cells) and for active pairs only. HHI is the Herfindahl–Hirschman index of each lender's lending concentration across industries. Active Pairs per Lender-Quarter is the average number of industries with positive lending per lender-quarter. Groups are defined by loan type (0=Express/line of credit, 1=Standard Guaranty/term loan) and post policy period.

Table 5: DID Regression Results: Lending Portfolio

	(1) $\lvert Spec \rvert$	(2) $Spec^2$	(3) HHI
Post × Treated	-0.00193**	-0.00104	0.0129*
	(0.0008)	(0.0008)	(0.007)
Quarter FE	Yes	Yes	Yes
Lender-Industry-Subprogram FE	Yes	Yes	Yes
Observations	470,133	470,133	43,406
Adjusted R^2	0.596	0.533	0.132
F	5.71	1.80	3.53

Note: This table reports regression results from equation (8). The dependent variables are the absolute and squared value of industry specialization ($Spec$) and the Herfindahl–Hirschman Index. The variable *Treated* indicates the treatment group (Standard Guaranty under the screening policy). The variable *Post* is a binary indicator equal to one in the post-intervention period. The specification includes lender-industry-subprogram fixed effects and quarter fixed effects, as indicated in the table. Standard errors clustered at the lender level are reported in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Decomposition: New Entries in Industries vs Portfolio Reallocation

	(1) $\lvert Spec \rvert$	(2) New Entry
Post × Treated	-0.0155***	0.0343***
	(0.0034)	(0.0072)
Quarter FE	Yes	Yes
Lender-Industry-Subprogram FE	Yes	Yes
Observations	82,472	470,133
Adjusted R^2	0.555	0.281
F	21.2	23.0

Note: Column (1) reports the *Portfolio Reallocation* effect on $\lvert Spec \rvert$, restricted to active lender-industry pairs ($L > 0$). Column (2) reports the *New Entry* effect using a Linear Probability Model where the dependent variable equals one if the lender has positive lending in the industry. *Treated* indicates Standard Guaranty loans; *Post* equals one after 2014:Q3. Both specifications include lender-industry-subprogram and quarter fixed effects. Standard errors clustered at the lender level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Specialization Decomposition with RCA3

	Full Sample (1) Absolute RCA3	Active Only (2) Absolute RCA3
Post × Treated	-0.00273**	-0.0262***
	(0.00115)	(0.00355)
Quarter FE	Yes	Yes
Lender-Industry-Subprogram FE	Yes	Yes
Observations	312,107	82,472
Adjusted R^2	0.184	0.449
F	5.698	54.51

Note: This table re-estimates the baseline DID specification from Eq. (8) using the unweighted portfolio deviation measure ($RCA3$) defined in Eq. (5) as the dependent variable. Unlike $Spec$ used in the main analysis, $RCA3$ isolates the internal portfolio-share deviation independently of the lender's total market share, capturing the pure portfolio-composition effect. Column (1) estimates the model on the full balanced panel including zero-lending cells; column (2) restricts to active lender–industry pairs with positive lending. Standard errors clustered at the lender level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Dynamic Panel (Two-Step System GMM): Specialization Persistence

	With Zero Investment		No Zero Investment†	
	(1) AR(1)	(2) AR(2)	(3) AR(1)	(4) AR(2)
$L_1(Spec)$	0.9747***	1.0889***	0.9500***	0.4030*
	(0.0140)	(0.3744)	(0.0330)	(0.2282)
$L_2(Spec)$		-0.1131		0.5950***
		(0.3748)		(0.2147)
$L_1(Spec) \times$ Post	-0.0545***	-0.4412	-0.1319***	0.1102
	(0.0153)	(0.3904)	(0.0252)	(0.2722)
$L_2(Spec) \times$ Post		0.4095		-0.2448
		(0.3868)		(0.2524)
Long-run persistence (pre)	0.975	0.976	0.950	0.998
Long-run persistence (post)	0.920	0.944	0.818	0.863
Observations	456,591	443,049	38,349	25,940
Groups	13,542	13,542	5,924	3,139
Instruments	57	76	57	107
AR(1) p-value	0.000	0.002	0.000	0.000
AR(2) p-value	0.000	0.407	0.000	0.111
AR(3) p-value	0.239	0.904	0.262	0.105
Hansen J p-value	0.083	0.050	0.096	0.164

Note: Two-step System GMM with Windmeijer-corrected standard errors and collapsed instrument sets. Columns (1) and (2) estimate on the full balanced panel (including zero-lending cells); columns (3) and (4) restrict to active lender–industry panels with positive lending. Columns (1) and (3) estimate an AR(1) model using instruments from lag(2∼6); columns (2) and (4) estimate an AR(2) model using instruments from lag(3∼6). Long-run persistence is the sum of lag coefficients; long-run persistence (post) adds the post-interaction terms. AR(k) are Arellano–Bond tests for serial correlation in differenced residuals. Hansen J tests overidentifying restrictions (robust to heteroskedasticity). †The active-only specifications (columns 3–4) pass Hansen J ($p = 0.096$ and 0.164). Three of four specifications comfortably pass Hansen J at the 5% level; column (2) is borderline ($p = 0.050$). The AR(2) test rejects for the AR(1) specifications ($p < 0.001$ in columns 1 and 3), indicating that lag(2) instruments may be invalid; the preferred AR(2) specifications (columns 2 and 4) address this by instrumenting from lag(3) onward and pass the AR(2) test ($p = 0.407$ and 0.111).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: DID Regression Results: Screening/Monitoring, Heterogeneous Effects, and New Borrowers

	(1) *Chgoff*	(2) *Relationship*	(3) *Spread*
Panel A: Baseline			
Post × Treated	-0.0098***	0.0025	-0.219***
	(0.003)	(0.006)	(0.041)
$\overline{Spec}$	-0.0139**	0.0595***	-0.127
	(0.007)	(0.021)	(0.109)
Adjusted R^2	0.109	0.102	0.597
F	79.6	35.9	163.7
Observations	292,794	292,794	292,794
Panel B: Heterogeneous Effect (Interaction with $\overline{Spec}$)			
Post × Treated	-0.0109***	0.0016	-0.208***
	(0.004)	(0.006)	(0.041)
Post × Treated × $\overline{Spec}$	0.0366**	-0.0236	-0.163
	(0.018)	(0.028)	(0.290)
Adjusted R^2	0.109	0.102	0.597
F	63.6	27.0	131.6
Observations	292,794	292,794	292,794
Panel C: New Borrower Interaction			
Post × Treated	0.0051	–	-0.125***
	(0.006)		(0.044)
Post × Treated × New Borrower	-0.0182***	–	-0.111***
	(0.006)		(0.033)
Adjusted R^2	0.109	–	0.597
F	62.4	–	125.2
Observations	292,794	–	292,794

Notes: Table reports results from Eq. (9). Entries are coefficients with standard errors in parentheses. $\overline{Spec}$ is the pre-policy median specialization for each lender–industry–loan-type cell. Fixed effects (lender-time, industry-time, state-time) and loan controls are the same across Panels A and B; Panel C does not condition on pre-policy specialization ($\overline{Spec}$) but uses the same sample as Panels A and B (292,794 observations). Standard errors are clustered at the lender–industry–subprogram level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Specialization Effect

	(1) *Chgoff*	(2) *Relationship*	(3) *Spread*
$Spec$	-0.0107***	0.0455***	-0.125**
	(0.002)	(0.008)	(0.049)
Lender-Time FE	Yes	Yes	Yes
Industry-Time FE	Yes	Yes	Yes
State-Time FE	Yes	Yes	Yes
Subprogram-Time FE	Yes	Yes	Yes
Other Loan Controls	Yes	Yes	Yes
Observations	293,657	293,657	293,657
Adjusted R^2	0.109	0.103	0.596
F	95.57	48.84	178.6

Note: Table reports results from Eq. (10). The dependent variables are default, relationship lending, and loan spread. $Spec$ is the specialization index (2), scaled by 100. The time frequency is quarterly. Covariates include loan terms (log loan amount, maturity in months, fixed vs. floating rate, guaranty ratio) and borrower characteristics (business type, franchise indicator, collateral). The specification includes fixed effects for subprogram-by-time, lender-by-time, industry-by-time, and state-by-time. Standard errors in parentheses, clustered at the lender–industry level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: First-Stage Results (Bartik IV for Specialization)

	(1) *Chgoff*	(2) *Relationship*	(3) *Spread*
Instrument effect on Spec:	(common first stage)		
Kleibergen–Paap Wald F	38.82	38.75	38.65
Kleibergen–Paap LM χ^2	8.96	8.88	8.79
Observations	193,727	193,727	193,727

Notes: Outcome columns label the corresponding second-stage dependent variable; all first stages regress $Spec$ on the Bartik instrument and the same control set used in the second stage. Standard errors and F/LM statistics are robust to heteroskedasticity and clustering by lender–industry level. Stock–Yogo weak-ID 10% critical value for a single endogenous regressor is 16.38; all Kleibergen–Paap F statistics exceed this threshold. Sample restricted to post-credit-scoring period (2014:Q3 onward). Specifications include high-dimensional fixed effects (lender×time, industry×time, state×time, type×time, plus firm-type indicators). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Specialization Effect: Bartik IV Estimation

	(1) *Chgoff*	(2) *Relationship*	(3) *Spread*
Panel A: IV Baseline			
$Spec$	-0.0123***	0.0650***	-0.116
	(0.004)	(0.007)	(0.072)
Lender-Time FE	Yes	Yes	Yes
Industry-Time FE	Yes	Yes	Yes
State-Time FE	Yes	Yes	Yes
Other Loan Controls	Yes	Yes	Yes
Observations	193,727	193,727	193,727
Adjusted R^2	0.051	-0.026	0.194
KP Wald F	38.82	38.75	38.65
Panel B: IV with New Borrower Interaction			
$Spec$	-0.0185***	—	-0.224***
	(0.005)		(0.056)
$Spec \times$ New Borrower	0.0074	—	0.132*
	(0.006)		(0.072)
Lender-Time FE	Yes	—	Yes
Industry-Time FE	Yes	—	Yes
State-Time FE	Yes	—	Yes
Other Loan Controls	Yes	—	Yes
Observations	193,727	—	193,727
Adjusted R^2	0.051	—	0.190
KP Wald F	13.59	—	13.52

Notes: Table reports results from Eq. (10) with two-stage least squares (Bartik IV). Sample restricted to post-credit-scoring period (2014:Q3 onward). The endogenous regressor is industry specialization ($Spec$). Panel A presents the baseline IV specification. Panel B adds a new-borrower interaction; both $Spec$ and $Spec \times$ New Borrower are instrumented with the Bartik IV and its interaction with the new-borrower indicator. The Bartik instrument is constructed from pre-period lender-by-industry shares interacted with leave-one-out national industry shifts; instrument components are computed on the full market before sample filters. KP Wald F = Kleibergen–Paap Wald F statistic; Stock–Yogo 10% critical values: 16.38 (single endogenous), 7.03 (two endogenous). Negative adjusted R^2 values are standard in 2SLS estimation because the objective function minimizes the IV moment conditions rather than the sum of squared residuals. Anderson–Rubin tests (robust to weak instruments): Panel B charge-off $\chi^2(2) = 7.21$, $p = 0.027$; Panel B spread $\chi^2(2) = 3.70$, $p = 0.157$. Standard errors clustered at lender–industry level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Rotemberg Weights: Bartik IV Decomposition by Industry

		Rotemberg		$\hat{\beta}_s$: Just-ID IV	
NAICS	Industry	$\hat{\alpha}_s$	Cum.%	Chgoff	Spread
48	Transportation	0.6558	65.6%	-0.0180	-0.1922
23	Construction	0.1099	76.6%	-0.0315	0.3924
54	Professional Svc	0.0839	85.0%	0.0036	-0.1377
62	Health Care	0.0832	93.3%	0.0367	-0.0503
72	Accommodation	0.0197	95.2%	0.0305	0.4043
42	Wholesale Trade	0.0190	97.2%	-0.0223	-0.4971
81	Other Services	0.0180	98.9%	-0.0340	-0.5241
52	Finance/Insurance	0.0030	99.2%	-0.0558	-0.0344
56	Admin/Waste Svc	0.0026	99.5%	0.0945	-0.0036
44	Retail Trade	0.0013	99.6%	-0.0248	-0.7931
53	Real Estate	0.0007	99.7%	-0.1663	0.7622
31	Manufacturing	0.0007	99.8%	-0.0637	0.5050
45	Retail Trade	0.0006	99.8%	-0.1775	-1.1803
71	Arts/Recreation	0.0005	99.9%	0.7252	-12.0317
61	Education	0.0004	99.9%	0.1157	-0.1490
51	Information	0.0003	100.0%	-0.0411	0.7650
33	Manufacturing	0.0002	100.0%	-0.5458	8.4682
32	Manufacturing	0.0002	100.0%	-0.1438	1.2398
11	Agriculture	0.0000	100.0%	3.3789	-1.0e+02
	Total	1.0000	100.0%	-0.0123	-0.1159

Weights decompose the Bartik IV estimate by NAICS 2-digit industry. $\hat{\beta}_{IV} = \sum_s \hat{\alpha}_s \hat{\beta}_s$. Industries sorted by $|\hat{\alpha}_s|$.

Notes: Table decomposes the Bartik IV estimate (Eq. (10)) into industry-specific contributions following Goldsmith-Pinkham et al. (2020). $\hat{\alpha}_s$ is the Rotemberg weight for NAICS 2-digit industry s; $\hat{\beta}_s$ is the just-identified IV estimate using only industry s's variation. The overall IV estimate satisfies $\hat{\beta}_{IV} = \sum_s \hat{\alpha}_s \hat{\beta}_s$. All weights are positive, ensuring the IV estimate is a proper weighted average of industry-specific effects. Industries sorted by $|\hat{\alpha}_s|$.

Table 14: Leave-One-Industry-Out: Bartik IV Robustness

	Chgoff		Relationship		Spread		
Dropped Industry	$\hat{\beta}$	s.e.	$\hat{\beta}$	s.e.	$\hat{\beta}$	s.e.	KP F
None (Baseline)	-0.0123	(0.0043)	0.0650	(0.0070)	-0.1158	(0.0723)	—
Agriculture	-0.0122	(0.0044)	0.0647	(0.0071)	-0.1134	(0.0721)	38.9
Construction	-0.0096	(0.0043)	0.0734	(0.0060)	-0.2196	(0.0457)	55.1
Manuf. (31)	-0.0121	(0.0043)	0.0647	(0.0070)	-0.1145	(0.0719)	38.9
Manuf. (32)	-0.0122	(0.0044)	0.0647	(0.0071)	-0.1149	(0.0720)	38.9
Manuf. (33)	-0.0121	(0.0044)	0.0646	(0.0071)	-0.1153	(0.0717)	39.0
Wholesale	-0.0121	(0.0043)	0.0651	(0.0069)	-0.1019	(0.0747)	38.5
Retail (44)	-0.0122	(0.0043)	0.0645	(0.0069)	-0.1112	(0.0731)	39.3
Retail (45)	-0.0119	(0.0043)	0.0646	(0.0069)	-0.1145	(0.0727)	39.1
Transportation	-0.0037	(0.0136)	0.0347	(0.0098)	0.0239	(0.1339)	172.7
Information	-0.0121	(0.0043)	0.0648	(0.0070)	-0.1152	(0.0725)	38.9
Finance/Ins.	-0.0119	(0.0044)	0.0652	(0.0069)	-0.1144	(0.0727)	38.7
Real Estate	-0.0121	(0.0045)	0.0647	(0.0070)	-0.1173	(0.0734)	38.9
Professional	-0.0134	(0.0048)	0.0672	(0.0070)	-0.1088	(0.0823)	34.9
Admin/Waste	-0.0128	(0.0043)	0.0660	(0.0071)	-0.1217	(0.0716)	39.2
Education	-0.0123	(0.0043)	0.0648	(0.0070)	-0.1151	(0.0723)	38.9
Health Care	-0.0160	(0.0035)	0.0672	(0.0066)	-0.1163	(0.0748)	36.2
Arts/Rec.	-0.0128	(0.0043)	0.0650	(0.0070)	-0.1062	(0.0724)	39.3
Accommodation	-0.0124	(0.0042)	0.0640	(0.0075)	-0.1239	(0.0686)	47.3
Other Svc.	-0.0114	(0.0046)	0.0643	(0.0067)	-0.1086	(0.0726)	38.8

Notes: Each row drops the named 2-digit NAICS industry and re-estimates Eq. (10) (Bartik IV, post-credit-scoring period). KP F = Kleibergen–Paap Wald F statistic for weak identification. Dropping Transportation (NAICS 48) substantially attenuates all coefficients, consistent with the Rotemberg weight concentration documented in Table 13. The instrument remains strong without Transportation ($F = 172.7$), but the second-stage estimates collapse, indicating that remaining industries provide directionally consistent but imprecise identifying variation.

Table 15: Right-Attrition Robustness: Charge-Off Regressions on Resolved Loans

	(1) *Chgoff*	(2) *Default (IV)*	(3) *Relationship*	(4) *Spread*
$Spec$	-0.0118***	-0.0149***	0.0452***	-0.1158**
	(0.003)	(0.004)	(0.008)	(0.047)
Lender-Time FE	Yes	Yes	Yes	Yes
Industry-Time FE	Yes	Yes	Yes	Yes
State-Time FE	Yes	Yes	Yes	Yes
Other Loan Controls	Yes	Yes	Yes	Yes
KP Wald F	—	42.43	—	—
Observations	266,812	168,619	266,812	266,812
Adjusted R^2	0.137	0.076	0.104	0.597
F	99.1	80.4	46.9	167.7

Notes: Sample restricted to loans with resolved status (charged-off, paid-in-full, or cancelled), excluding 26,461 unresolved (EXEMPT) loans (9.0% of baseline). Columns (1)–(2) report the OLS and IV charge-off specifications; columns (3)–(4) report OLS repeated-borrower and spread specifications. The IV first-stage Kleibergen–Paap F-statistic improves from 38.8 to 42.4 after dropping unresolved loans. Standard errors clustered at the lender $\times$ industry level.

Table 16: Seasoning Restriction Robustness: Loans Originated Before 2017:Q1

	(1) *Chgoff*	(2) *Default (IV)*	(3) *Relationship*	(4) *Spread*
Spec	-0.0065**	-0.0061	0.0352***	-0.1394***
	(0.003)	(0.005)	(0.008)	(0.051)
Lender-Time FE	Yes	Yes	Yes	Yes
Industry-Time FE	Yes	Yes	Yes	Yes
State-Time FE	Yes	Yes	Yes	Yes
Other Loan Controls	Yes	Yes	Yes	Yes
KP Wald F	—	47.97	—	—
Observations	196,312	96,382	196,312	196,312
Adjusted R^2	0.097	0.047	0.098	0.574
F	101.6	76.9	50.3	134.3

Notes: Sample restricted to loans originated before 2017:Q1, ensuring at least three years to reach a terminal status by the sample end (2019:Q4). This restriction reduces the unresolved (EXEMPT) share from 24.31% to 13.68% among post-period treated loans and from 8.18% to 2.49% among controls. Column (1): OLS charge-off. Column (2): IV charge-off (post-credit-scoring period only; Kleibergen–Paap $F = 48.0$). Column (3): OLS repeated-borrower flag. Column (4): OLS spread. All specifications include lender-time, industry-time, state-time, and subprogram-time fixed effects, with standard errors clustered at the lender $\times$ industry level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Matching Robustness: CEM and PSM Samples

	CEM			PSM		
	(1) *Chgoff*	(2) *Relationship*	(3) *Spread*	(4) *Chgoff*	(5) *Relationship*	(6) *Spread*
Panel A: DID Baseline						
Post × Treated	0.0108**	-0.0093	-0.2856***	-0.0285***	-0.0071	-0.1075**
	(0.005)	(0.009)	(0.046)	(0.006)	(0.009)	(0.049)
Observations	135,277	135,277	135,277	92,960	92,960	92,960
Adjusted R^2	0.097	0.128	0.609	0.114	0.105	0.529
Panel B: Triple Interaction (*Post* × *Treated* × $\overline{Spec}$)						
Post × Treated	-0.0016		-0.1784***	-0.0284***		-0.0172
	(0.005)		(0.046)	(0.009)		(0.052)
Post × Treated × $\overline{Spec}$	0.1099***		-0.8946***	0.0052		-0.7119***
	(0.024)		(0.177)	(0.038)		(0.259)
Observations	135,277		135,277	92,960		92,960
Adjusted R^2	0.098		0.610	0.115		0.530

Notes: Panel A re-estimates the DID baseline (Eq. (9)) on matched samples. Panel B re-estimates the triple interaction with pre-policy specialization. CEM: Coarsened Exact Matching on log loan amount, maturity, rate type, repeated-borrower status, collateral, revolver status, corporate type, and franchise status (135,277 obs). PSM: Propensity Score Matching using the same covariates with nearest-neighbor matching (92,960 obs). Guaranty ratio excluded because Express and Standard Guaranty loans have structurally different guarantee levels (50% vs. 75–85%). Both samples include lender-time, industry-time, and state-time fixed effects. Standard errors clustered at the lender–industry–subprogram level.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Internet Appendix

Internet Appendix for: "Public Information, Private Expertise, and Lender Specialization"

This Internet Appendix accompanies the main paper. Section references of the form "Section X" refer to the main paper; sections within this appendix are labeled IA.A through IA.H. Tables and figures are numbered IA.1, IA.2, etc.

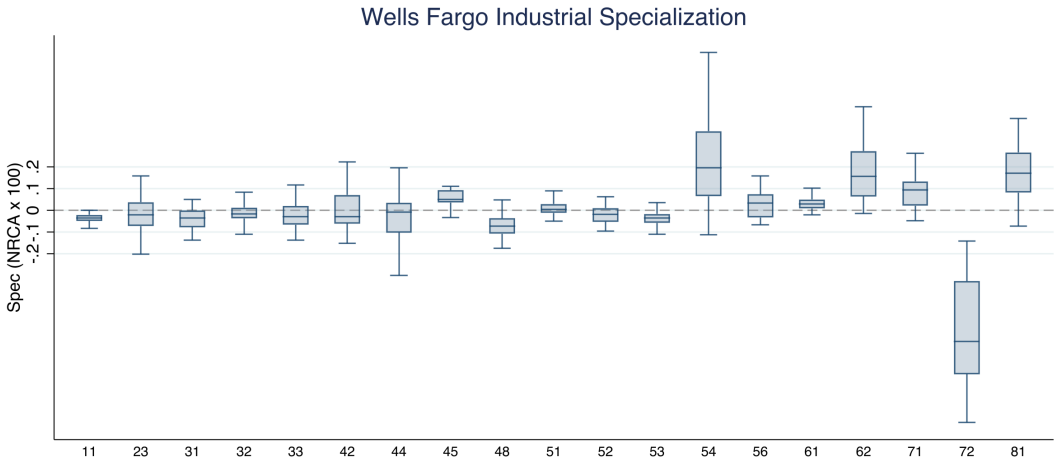


Figure IA.1: Wells Fargo Industry Specialization

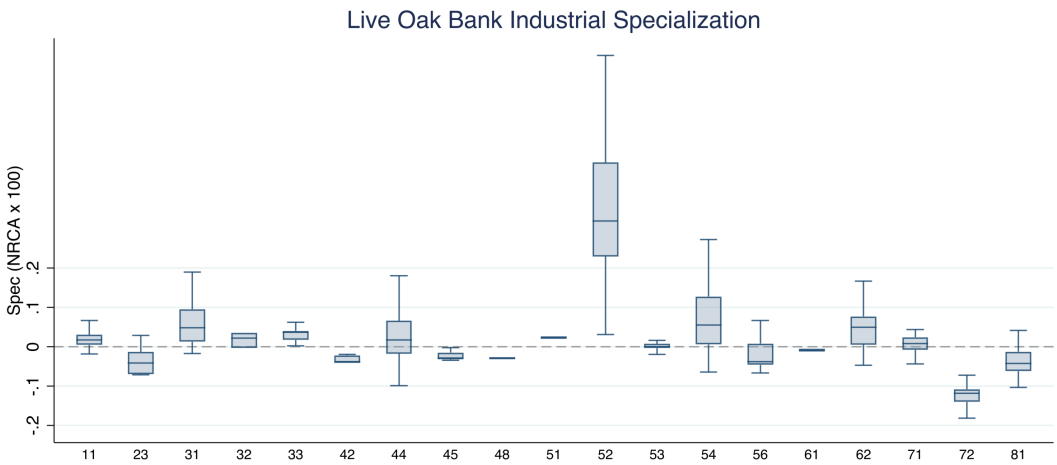


Figure IA.2: Live Oak Bank Industry Specialization

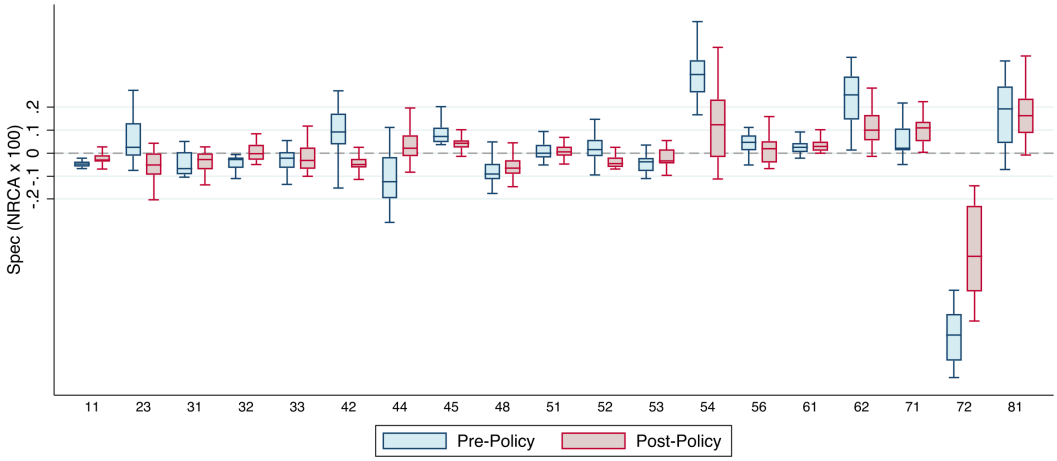


Figure IA.3: Wells Fargo Industry Specialization — Before and After 2014

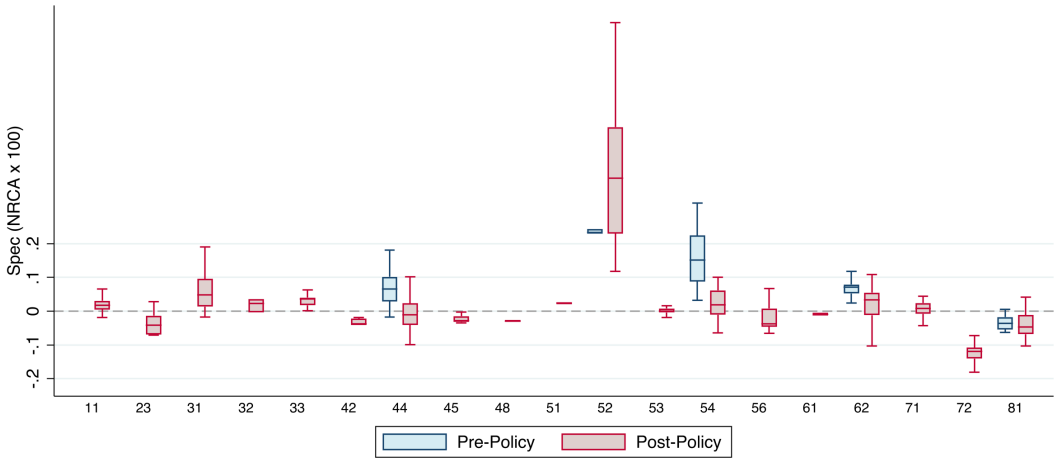


Figure IA.4: Live Oak Bank Industry Specialization — Before and After 2014

A Specialization Index

A proper specialization measure should satisfy two properties (Paravisini et al., 2023): it must capture a lender's comparative advantage in a particular industry rather than a general lender-wide advantage, and it must be comparable across industries and lenders regardless of size. This appendix defines three candidate indices, proves that two fail the comparability requirement, and shows that the NRCA index (denoted $Spec$ in the body) satisfies both properties.

A.1 Notation

Let B be the set of lenders and S the set of industries. Let $|B|$ denote the number of lenders. The time subscript t is suppressed for clarity.

$$L \equiv \sum_{b\in B}\sum_{k\in S} L_{bk} \qquad \text{(total market lending)} \tag{IA.1}$$

$$L_j \equiv \sum_{k\in S} L_{jk} \qquad \text{(lender } j\text{'s total lending)} \tag{IA.2}$$

$$L_s \equiv \sum_{j\in B} L_{js} \qquad \text{(industry } s\text{'s total lending)} \tag{IA.3}$$

A.2 Incompatibility of alternative measures

Balassa RCA ($RCA2$).

$$RCA2_{js} = \left(\frac{L_{js}}{\sum_{k\in S} L_{jk}}\right) \Big/ \left(\frac{\sum_{b\in B} L_{bs}}{\sum_{b\in B}\sum_{k\in S} L_{bk}}\right) = \frac{L_{js}}{L_j} \Big/ \frac{L_s}{L} = \frac{L \cdot L_{js}}{L_j \cdot L_s} \tag{IA.4}$$

Sum across lenders is non-constant. Since L_j varies with j:

$$\sum_{j\in B} RCA2_{js} = \sum_{j\in B} \frac{L \cdot L_{js}}{L_j \cdot L_s} = \frac{L}{L_s}\sum_{j\in B}\frac{L_{js}}{L_j} \tag{IA.5}$$

The term $\sum_{j\in B} L_{js}/L_j$ depends on how each lender allocates credit to industry s relative to its total lending, so the sum is non-constant across industries.

Sum across industries is non-constant. Since L_s varies with s:

$$\sum_{s\in S} RCA2_{js} = \sum_{s\in S} \frac{L \cdot L_{js}}{L_j \cdot L_s} = \frac{L}{L_j}\sum_{s\in S}\frac{L_{js}}{L_s} \tag{IA.6}$$

The term $\sum_{s\in S} L_{js}/L_s$ depends on lender j's credit share in each industry, so the sum

is non-constant across lenders.

Difference RCA ($RCA3$).

$$RCA3_{js} = \frac{L_{js}}{L_j} - \frac{L_s}{L} \tag{IA.7}$$

Sum across industries is zero:

$$\sum_{s \in S} RCA3_{js} = \sum_{s \in S} \left[\frac{L_{js}}{L_j} - \frac{L_s}{L} \right] = \frac{L_j}{L_j} - \frac{L}{L} = 0 \tag{IA.8}$$

Sum across lenders is non-zero. Since L_j varies with j and the constant L_s/L is summed $|B|$ times:

$$\sum_{j \in B} RCA3_{js} = \sum_{j \in B} \frac{L_{js}}{L_j} - |B| \cdot \frac{L_s}{L} \tag{IA.9}$$

Lenders with smaller L_j contribute disproportionately to $\sum L_{js}/L_j$, introducing a small-lender bias. Unless every lender allocates to industry s in proportion to the market average, $RCA3$ is not comparable for cross-lender analyses.

A.3 Compatibility of $Spec$ ($\equiv$ NRCA)

The NRCA index, denoted $Spec$ in the body, corresponds to RCA in Yu et al. (2009):

$$RCA_{js} = \frac{L_{js} - \frac{L_s \cdot L_j}{L}}{L} = \frac{L_{js}}{L} - \frac{L_s}{L} \cdot \frac{L_j}{L} \tag{IA.10}$$

Sum across lenders is zero:

$$\sum_{j \in B} RCA_{js} = \sum_{j \in B} \left[\frac{L_{js}}{L} - \frac{L_s}{L} \cdot \frac{L_j}{L} \right] = \frac{L_s}{L} - \frac{L_s}{L} \cdot \frac{L}{L} = 0 \tag{IA.11}$$

Sum across industries is zero:

$$\sum_{s \in S} RCA_{js} = \sum_{s \in S} \left[\frac{L_{js}}{L} - \frac{L_s}{L} \cdot \frac{L_j}{L} \right] = \frac{L_j}{L} - \frac{L}{L} \cdot \frac{L_j}{L} = 0 \tag{IA.12}$$

Because $Spec$ sums to zero across both lenders and industries, it is comparable for cross-

lender and cross-industry analyses without the size-dependent biases that affect $RCA2$ and $RCA3$.

B Borrower Identification and Relationship Lending Measurement

This appendix describes the construction of borrower identities and the relationship lending indicator used throughout the paper. Because the SBA 7(a) data do not contain a unique, time-consistent firm identifier, borrowers must be linked across loan records using observable fields. The procedure has three stages: field standardization, fuzzy record linkage to assign a borrower identifier, and construction of relationship lending and prior-SBA indicators from the linked panel.

B.1 Field Standardization

Before any matching is performed, three fields—borrower name, street address, and city—are cleaned and standardized to maximize the precision of downstream comparisons.

Borrower name. Raw borrower names are converted to uppercase and stripped of non-ASCII characters and excess whitespace. Placeholder names (e.g., "TBD," "UNKNOWN," "N/A") are flagged and set to missing. Legal names are separated from trade names using "DBA," "D/B/A," and related delimiters; for example, `ACME INC DBA BEST PIZZA` yields a legal name of `ACME INC` and a DBA of `BEST PIZZA`. Business entity suffixes are mapped to canonical short forms to eliminate variation in abbreviation and punctuation (e.g., `LIMITED LIABILITY COMPANY`, `L.L.C.`, and `LLC` all map to `LLC`), including truncations caused by SBA field-length limits (`INCORPORAT` → `INC`). Trailing location identifiers such as `UNIT 5`, `STE 101`, and `STORE #23` are stripped to prevent distinct establishments of the same firm from appearing as separate borrowers.

Street address. Raw addresses are converted to uppercase and cleaned in several steps: typographic corrections (e.g., dollar signs used as hash marks), PO Box normalization, secondary unit extraction into a separate field, directional standardization (`NORTH` → `N`), street type standardization to USPS abbreviations (`STREET` → `ST`, `AVENUE` → `AVE`), and street number extraction. The output comprises a standardized street string, a separate street number and street name, a standardized unit string, and a PO Box indicator.

City. City names are converted to uppercase, stripped of parenthetical annotations and embedded state abbreviations, and standardized for common prefixes (`SAINT` $\rightarrow$ `ST`, `FORT` $\rightarrow$ `FT`, `MOUNT` $\rightarrow$ `MT`).

B.2 Fuzzy Record Linkage

Borrower identities are assigned by linking loan records that refer to the same firm using a blocking-and-matching algorithm implemented with a Union-Find (disjoint set) data structure.

Blocking. Because all-pairs comparison across N loan records is computationally infeasible ($\binom{N}{2}$ pairs), candidate pairs are restricted to records sharing at least one blocking key. Three complementary strategies are employed: (i) state $\times$ ZIP $\times$ first 12 characters of the standardized name; (ii) state $\times$ city $\times$ first 12 characters of the standardized name; and (iii) state $\times$ ZIP $\times$ street number $\times$ first 14 characters of the standardized street name. Blocks exceeding 700 records are skipped, and within-block comparisons are capped at 300,000 pairs.

Similarity metric. All string comparisons use the *token set ratio*, which is based on the Levenshtein edit distance. The metric tokenizes each string, computes intersection and residual token sets, and returns the maximum normalized similarity (0–100 scale) among three pairwise comparisons of sorted token combinations. This metric is robust to token reordering (e.g., `SMITH JOHN LLC` vs. `JOHN SMITH LLC`) and partial token overlap.

Matching passes. The algorithm proceeds in three passes:

1. *Strong matches.* Records are linked if they satisfy any of: (a) exact address match and name similarity ≥ 88; (b) same ZIP, same street number, street similarity ≥ 92, and name similarity ≥ 88; or (c) same ZIP, street similarity ≥ 92, and name similarity ≥ 92. Records with generic names (fewer than two tokens or seven characters in the non-suffix core) require criterion (a) with name similarity ≥ 92. Records with conflicting franchise codes are linked only under criterion (a) with an exact address match, preventing distinct franchise locations from being collapsed.

2. *Weak matches.* Remaining unlinked pairs are linked if: (a) name similarity ≥ 96 and same ZIP or city; or (b) name similarity ≥ 92, street similarity ≥ 86, and same ZIP or city. Generic names are excluded, and a merge guard prevents unions across states or spanning more than five distinct ZIP codes.

3. *DBA cross-match.* Records are linked where one record's trade name exactly matches another's legal name within the same city and state, subject to the same merge guards.

After all three passes, the connected components of the Union-Find structure define the borrower identities. Each component is assigned a unique borrower identifier determined by the minimum SBA loan number within the component, ensuring deterministic and reproducible assignment.

B.3 Relationship Lending and Prior-SBA Indicators

Two indicators are constructed from the linked panel using each loan's approval date, lender identifier, and the newly assigned borrower identifier.

Relationship lending. A loan is classified as *relationship lending* if the borrower has obtained at least one prior SBA loan from the *same lender*. Formally, let $\mathcal{L}_{b,\ell}(t)$ denote the set of loans originated by lender ℓ to borrower b with approval dates strictly before t. Then

$$\text{RelationshipLending}_i = \mathbf{1}\left[\left|\mathcal{L}_{b(i),\,\ell(i)}(t_i)\right| > 0\right] \tag{IA.13}$$

where $b(i)$, $\ell(i)$, and t_i denote loan i's borrower, lender, and approval date. This definition captures repeat borrowing from the same institution, which is the standard proxy for relationship lending in the literature. For loans originated on the same date by the same lender to the same borrower, none is classified as a relationship loan, because temporal precedence cannot be established.

Previous SBA borrower. A loan is classified as *previous SBA* if the borrower has obtained at least one prior SBA loan from *any* lender:

$$\text{PreviousSBA}_i = \mathbf{1}\left[\left|\bigcup_{\ell' \in B} \mathcal{L}_{b(i),\,\ell'}(t_i)\right| > 0\right] \tag{IA.14}$$

where B is the set of all lenders. This indicator captures whether the borrower has any prior SBA financing experience, regardless of the originating institution.

B.4 Measurement Limitations

False positives. The multi-threshold design mitigates the risk of incorrectly linking distinct borrowers: strong matches require both name and address agreement, weak matches demand very high name similarity (≥ 96) or the combination of high similarity (≥ 92) with address support, and the merge guard prevents geographically implausible unions. The generic-name filter prevents high-frequency short names from generating false links, and franchise-code conflicts are treated as presumptive evidence of distinct establishments unless an exact address match is present.

False negatives. Records with missing or erroneous ZIP codes, city names, or highly dissimilar name spellings may fail to enter any common block and thus remain unlinked. The three complementary blocking strategies provide multiple paths for a true pair to enter candidate comparison, and the DBA cross-match pass recovers links that would otherwise be missed when a firm operates under a trade name that differs substantially from its legal name.

Temporal structure. The relationship lending indicator requires strict temporal precedence: a loan is classified as relationship lending only if the borrower has a prior loan from the same lender with an earlier approval date. Loans approved on the same date are not classified as relationship loans, producing a conservative measure. The borrower history used for classification extends to the inception of the SBA dataset (1991), providing a long lookback window that minimizes left-censoring of relationship histories.

C Bartik-Style Instrument Construction

This appendix constructs a Bartik (shift-share) instrument for the specialization index ($Spec$). Specialization is endogenous because lenders allocate credit to industries where they have expertise and tilt toward industries with better current or expected returns, so specialization co-moves with loan outcomes through lender selection.

C.1 Motivation and Construction

The share of lender j's investment in industry s relative to total market investment decomposes into a common industry–time shift and the lender's within-industry share:

$$\frac{L_{j,s,t}}{L_t} = \underbrace{\frac{L_{s,t}}{L_t}}_{\text{common industry–time shift}} \times \underbrace{\frac{L_{j,s,t}}{L_{s,t}}}_{\text{lender } j\text{'s within-industry share}} . \tag{IA.15}$$

The first term is plausibly exogenous to any individual lender's decision, while the second is endogenously determined by the lender's knowledge about industry s. Define:

$$g_{s,t} \equiv \frac{L_{s,t}}{L_t}, \qquad \text{(industry } s\text{'s share of total credit at } t) \tag{IA.16}$$

$$z_{j,s,t} \equiv \frac{L_{j,s,t}}{L_{s,t}}, \qquad \text{(lender } j\text{'s share within industry } s \text{ at } t) \tag{IA.17}$$

$$\omega_{j,t} \equiv \frac{L_{j,t}}{L_t}. \qquad \text{(lender } j\text{'s share of total credit at } t) \tag{IA.18}$$

The specialization measure from Section A can then be written as

$$Spec_{j,s,t} = \frac{L_{j,s,t}}{L_t} - \frac{L_{s,t}}{L_t} \cdot \frac{L_{j,t}}{L_t} = g_{s,t}\left(z_{j,s,t} - \omega_{j,t}\right). \tag{IA.19}$$

The endogeneity of $Spec$ arises from contemporaneous, lender-specific shocks captured by $z_{j,s,t}$ and $\omega_{j,t}$. The instrument fixes each lender's industry exposure at a pre-policy baseline t_0 and interacts it with a contemporaneous leave-one-out national industry shift $g_{s,t}^{(-j)}$, mapping common national reallocations into predicted, lender-specific movements in specialization. Figure IA.5 summarizes the identification logic.

Formally, the Bartik IV for $Spec$ is

$$Z_{j,s,t}^{\text{Bartik}} = g_{s,t}^{(-j)} \left(z_{j,s,t_0} - \omega_{j,t_0} \right) \tag{IA.20}$$

where $g_{s,t}^{(-j)} \equiv \frac{L_{s,t} - L_{j,s,t}}{L_t - L_{j,t}}$ is the leave-one-out national industry share that excludes lender j's own lending, mitigating mechanical correlation between the instrument and j's outcomes.

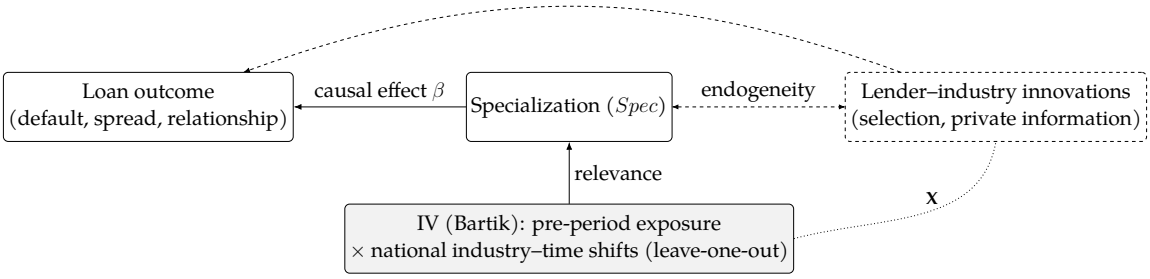


Figure IA.5: Bartik (shift-share) IV identification diagram. Solid arrows denote causal links; dashed arrows denote endogenous relationships; the dotted line marked **X** denotes the maintained exclusion restriction: conditional on the included fixed effects and controls, the national industry–time shifts used to form the instrument are orthogonal to lender–industry outcome innovations.

C.2 Identification Conditions

The instrumental variables approach relies on three conditions.

Exclusion restriction. The national industry–time shifts $g_{s,t}^{(-j)}$ must be orthogonal to lender–industry innovations in loan outcomes, conditional on the fixed effects and controls in the second-stage equation. Lender–time fixed effects absorb lender-wide time-varying shocks (balance-sheet capacity, risk appetite), industry–time fixed effects absorb industry-wide conditions (sector profitability, risk), and state–time fixed effects absorb local macroeconomic conditions. Under this conditioning set, identifying variation comes from common national reallocations interacted with predetermined lender exposures, not from contemporaneous idiosyncratic adjustments.

Within the Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2025) frameworks, three features support this restriction: (i) identification relies on exogeneity of pre-period exposure shares rather than the national shifts themselves; (ii) the leave-one-out

construction ensures j's own lending does not mechanically influence $g_{s,t}^{(-j)}$; and (iii) all 19 Rotemberg weights are positive, so the IV estimate is a proper convex combination without sign reversals.

The most plausible remaining threat is industry-specific credit-risk shocks that simultaneously shift the national industry lending share and directly worsen borrower outcomes. Industry–time fixed effects absorb shocks common to all lenders within an industry–quarter cell. For this threat to violate exclusion, it must operate through the *interaction* of the national shift with lender-specific pre-period exposure—that is, a lender's historical footprint must predict how an industry credit-risk shock differentially affects *that lender's* outcomes, beyond what the fixed effects capture. One such channel would be if specialized lenders hold loans with systematically different risk characteristics within the same industry–quarter cell. The loan-level controls (loan size, maturity, guarantee ratio, collateral, rate type) absorb much of this within-cell variation. A residual correlation would need to operate through channels orthogonal to both observable loan characteristics and the lender–time, industry–time, and state–time fixed effects—a restrictive requirement.

Relevance. The instrument must strongly predict specialization. This holds when lenders with larger pre-policy exposure to an industry adjust their specialization more in response to national shifts in that industry's share. The first-stage estimates confirm a strong, precisely estimated relationship (Kleibergen–Paap $F = 38$–39), alleviating weak-instrument concerns.

Monotonicity. Interpreting the IV as a local average treatment effect requires no systematic sign reversals in how specialization responds to the instrument. Economically, when an industry's national lending share expands, lenders with greater baseline exposure face lower marginal costs of scaling up, making a positive response more likely. The Rotemberg decomposition confirms all 19 weights are positive. The weight concentration and leave-one-industry-out results are discussed in Section 5 of the main paper; Section D of this appendix provides an alternative specification that addresses the resulting sensitivity.

D Alternative Shift-Share IV: Express-Derived Shifts

The baseline Bartik IV pools Standard Guaranty and Express lending in the shift component. Because Standard loans dominate aggregate SBA volume and their industry distribution is directly affected by the 2014 reform, these shifts may partially reflect policy-induced reallocation rather than exogenous demand variation—a concern amplified by Transportation's 65.6% Rotemberg weight.

To address this potential circularity, I construct an alternative instrument using shifts derived exclusively from Express lending. Because Express loans were exempt from the 2014 credit-scoring reform, Express industry shares reflect organic demand variation rather than policy-induced reallocation. Formally, the Express-shift IV replaces the pooled leave-one-out shift $g_{s,t}^{(-j)}$ with the Express-only national industry share $g_{s,t}^{\text{Express}}$, while retaining loan-type-specific pre-period weights:

$$Z_{j,s,t}^{\text{Express}} = g_{s,t}^{\text{Express}} \left(z_{j,s,t_0} - \omega_{j,t_0} \right). \tag{IA.21}$$

A leave-one-out correction is unnecessary because Standard loans do not enter the Express aggregates, so there is no mechanical correlation between a Standard lender's outcomes and the Express-derived shift. This instrument is applied to the Standard-only sample in the post-credit-scoring period, so that identification comes from Express-driven industry demand variation interacted with Standard lenders' predetermined exposure.

Feasibility. Express lending provides sufficient variation: the median Express share of total SBA lending across industry–quarter cells is 52.2%, with 100% cell coverage (no missing industry–quarter observations). The Pearson correlation between Express and Standard industry shares is $\rho = 0.614$, indicating that the two programs share common demand factors while differing enough to provide independent variation.

Exclusion restriction. The Express-shift IV requires that national reallocations of Express lending affect Standard borrower outcomes only through their effect on Standard lender specialization, conditional on the included fixed effects. Three features support this restriction. First, Express and Standard serve distinct borrower segments: Express carries a 50% guarantee (versus 75–85%), shorter maturities, and a higher share of lines of credit, limiting direct competition for borrowers across programs. Second,

the shift is computed at the national level while outcomes are at the loan level, with lender–time, industry–time, and state–time fixed effects absorbing local and aggregate demand shocks. Under the Borusyak et al. (2025) framework, the identifying assumption is exogeneity of national Express shifts conditional on pre-period exposure. Third, the flat post-reform trajectory of Express volume mitigates concerns about strategic substitution that could generate a direct Express-to-Standard spillover channel. A remaining threat is that Express and Standard borrower pools overlap within industries, so that an Express demand shock could alter Standard borrower composition through selection. The moderate correlation ($\rho = 0.614$) suggests meaningful but incomplete overlap.

Results. Table IA.1 presents four specifications for the charge-off outcome: (1) OLS on the full post-credit sample, (2) the baseline IV with pooled leave-one-out shifts, (3) the IV on the Express-only sample with Express shifts, and (4) the Express-shift IV on the Standard sample. Columns (1)–(3) agree on sign and magnitude: specialization reduces default (-0.011 OLS, -0.012 IV Pooled, -0.022 IV Express). Column (4) reverses sign ($+0.056$, $p = 0.027$), with a marginal first stage (Kleibergen–Paap $F = 13.2$, above the Stock–Yogo 15% critical value of 8.96 but below the 10% critical value of 16.38). Anderson–Rubin confidence intervals, which are robust to weak instruments, yield a 95% CI of $[-0.007, 0.121]$ for charge-off ($p_{\text{AR}} = 0.070$), including zero. For repeated borrower, the Express-shift IV confirms the baseline ($+0.065$, $p = 0.091$; AR 95% CI excludes zero: $[0.003, +\infty)$, $p_{\text{AR}} = 0.048$). For spread, the coefficient is directionally consistent but insignificant (-0.078, $p = 0.247$; AR $p = 0.148$).

Interpretation: industry heterogeneity. The charge-off divergence is driven by a complete redistribution of Rotemberg weights across the two IV specifications. Table IA.2 reports the Express-shift Rotemberg decomposition. Under pooled shifts, Transportation accounts for 65.6% of the identifying weight; under Express shifts, Transportation drops to 1.1%, while Accommodation (35.5%), Other Services (25.4%), and Health Care (14.7%) dominate. All 19 Express-shift weights are positive (no sign reversals), confirming that the Express-shift IV estimate is a proper weighted average of industry-specific effects. Section 5 of the main paper provides the full economic interpretation of the weight redistribution and the industry-heterogeneity channel.

Table IA.1: IV Comparison: Alternative Shift-Share Specifications (Charge-off)

	(1) *OLS*	(2) *IV Pooled*	(3) *IV Express*	(4) *IV Std+Expr*
Spec	-0.0114***	-0.0123***	-0.0217***	0.0562**
	(0.002)	(0.004)	(0.003)	(0.025)
Sample	Both	Both	Express	Standard
Instrument	—	Pooled LOO	Pooled LOO	Express non-LOO
KP Wald F	—	38.8	46.0	13.2
Adj. R^2	0.122	0.051	0.137	-0.060
Observations	193,727	193,727	138,343	53,418
Lender-Time FE	Yes	Yes	Yes	Yes
Industry-Time FE	Yes	Yes	Yes	Yes
State-Time FE	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes

Outcome: charge-off. Col. (4) uses Express industry shifts applied to Standard sample.

Notes: Outcome is charge-off (binary). All specifications include lender–time, industry–time, and state–time fixed effects, plus loan-level controls (spread, log loan amount, rate type, maturity, guaranty ratio, repeated borrower, collateral, revolver status). Columns (1)–(2) pool Standard and Express; column (3) restricts to Express loans with Express LOO shifts; column (4) restricts to Standard loans with Express non-LOO shifts. Standard errors clustered at the lender $\times$ industry level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.2: Rotemberg Weights: Express-Shift Bartik IV

		Rotemberg		$\hat{\beta}_s$: Just-ID IV	
NAICS	Industry	$\hat{\alpha}_s$	Cum.%	Chgoff	Spread
72	Accommodation	0.3555	35.5%	0.0524	0.0608
81	Other Services	0.2535	60.9%	-0.0862	0.0172
62	Health Care	0.1472	75.6%	0.1902	-0.4804
54	Professional Svc	0.1039	86.0%	0.3283	-0.2187
52	Finance/Insurance	0.0444	90.4%	-0.0123	0.4007
44	Retail Trade	0.0272	93.2%	-0.2319	-0.2516
56	Admin/Waste Svc	0.0175	94.9%	0.1081	0.2398
48	Transportation	0.0107	96.0%	0.2495	-0.2425
42	Wholesale Trade	0.0101	97.0%	0.3718	-0.3192
71	Arts/Recreation	0.0090	97.9%	0.1424	-0.6242
23	Construction	0.0074	98.6%	-0.1131	-1.0629
11	Agriculture	0.0055	99.2%	-0.2226	-0.7786
33	Manufacturing	0.0032	99.5%	-0.5557	-0.3295
31	Manufacturing	0.0029	99.8%	0.3166	1.3078
53	Real Estate	0.0008	99.9%	0.0375	-2.5467
51	Information	0.0004	99.9%	-1.1921	-1.0457
32	Manufacturing	0.0003	100.0%	0.3413	-4.4035
45	Retail Trade	0.0003	100.0%	-10.8564	-0.1279
61	Education	0.0002	100.0%	4.4678	-2.0122
	Total	1.0000	100.0%	0.0562	-0.0775

Standard sample. Express industry shifts as instrument.
$\hat{\beta}_{IV} = \sum_s \hat{\alpha}_s \hat{\beta}_s$. Industries sorted by $|\hat{\alpha}_s|$.

Notes: Standard sample, post-credit-scoring period. Instrument: Express industry shares $\times$ Standard pre-period weights. Industries sorted by $|\hat{\alpha}_s|$. $\hat{\beta}_{IV} = \sum_s \hat{\alpha}_s \hat{\beta}_s$.

E CSDID Robustness for Extensive-Margin DID

Table IA.3 compares the TWFE estimates from Eq. (8) of the main paper with the Callaway and Sant'Anna (2021) doubly-robust inverse-probability-weighted (DRIPW) estimator. The CSDID estimator is robust to heterogeneous treatment effects across cohorts, which could bias the TWFE interaction coefficient when treatment effects evolve over time.

For $|Spec|$, the CSDID post-treatment average (-0.00132, $p < 0.01$) confirms the direction and statistical significance of the TWFE estimate (-0.00193, $p = 0.017$). The smaller CSDID magnitude is consistent with the TWFE estimate placing more weight on later periods where the treatment effect grows. For $Spec^2$, the CSDID post-average (-0.00033, $p = 0.084$) is marginally significant, compared with the imprecise TWFE estimate (-0.00104, $p = 0.180$). For HHI, both estimators produce positive but imprecise estimates ($+0.013$ TWFE, $p = 0.061$; $+0.021$ CSDID, $p = 0.285$), directionally consistent with increased within-portfolio concentration. Pre-trend tests for all three outcomes are reported in Table IA.4.

Table IA.3: TWFE vs. CSDID Estimates: Extensive-Margin DID

	$\|Spec\|$		$Spec^2$		HHI	
	TWFE	CSDID	TWFE	CSDID	TWFE	CSDID
Post × Treated	-0.00193^{**}	-0.00132^{***}	-0.00104	-0.00033^{*}	$+0.01294^{*}$	$+0.02068$
	(0.00081)	(0.00033)	(0.00077)	(0.00019)	(0.00688)	(0.01935)
Observations	470,133	470,133	470,133	470,133	43,406	43,406

Notes: TWFE columns report the $Treated \times Post$ interaction with lender-industry-loan-type and quarter fixed effects. CSDID columns report the Callaway and Sant'Anna (2021) doubly-robust inverse-probability-weighted (DRIPW) estimator with never-treated as control group; the reported coefficient is the average of all post-treatment group-time ATTs. Both estimators cluster standard errors at the lender level. See Table IA.4 for pre-trend test statistics.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

F Parallel Trend Tests

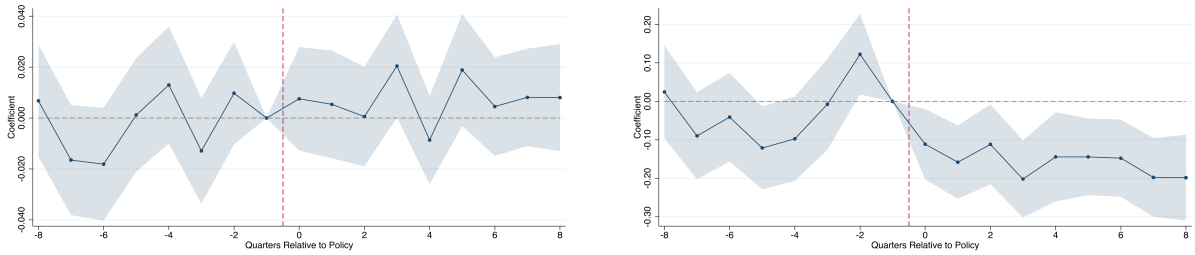
Table IA.4 reports the Callaway and Sant'Anna (2021) CSDID pre-trend test within an eight-quarter window preceding the policy, which jointly tests that all pre-treatment group-time average treatment effects equal zero. Standard errors are clustered at the lender level for comparability with the TWFE estimates. For $|Spec|$, the joint test fails to reject ($\chi^2(7) = 10.71$, $p = 0.152$), as do $Spec^2$ ($\chi^2(7) = 6.74$, $p = 0.457$) and HHI ($\chi^2(7) = 3.47$, $p = 0.838$). These results support the parallel-trends assumption for the extensive-margin specialization outcomes.

Figure IA.6 presents TWFE event-study plots for the loan-outcome DID, replacing the single $Treated \times Post$ interaction with quarter-by-treatment indicators ($t = -8$ to $t = 8$, with endpoints binned and $t = -1$ as reference). For charge-off rates (panel a), the pre-treatment coefficients oscillate tightly around zero with no individual coefficient statistically significant, providing visual confirmation of parallel pre-trends. For spreads (panel b), some pre-treatment coefficients deviate from zero but without a monotonic pattern: the deviations at $t = -5$ and $t = -2$ have opposite signs, indicating transitory noise rather than a confounding trend. The post-treatment coefficients for spreads show a clear and persistent negative shift, consistent with the policy reducing interest-rate spreads.

Table IA.4: Parallel Trend Tests: Extensive-Margin DID

	$\|Spec\|$	$Spec^2$	HHI
$\chi^2(7)$	10.71	6.74	3.47
p-value	[0.152]	[0.457]	[0.838]
Window	$(-8, 8)$	$(-8, 8)$	$(-8, 8)$

Notes: Table reports the Callaway and Sant'Anna (2021) CSDID joint pre-trend test (`estat pretrend`) within an eight-quarter window preceding the policy, which tests that all pre-treatment group-time average treatment effects equal zero under the doubly-robust inverse-probability-weighted (DRIPW) estimator. The $|Spec|$ and $Spec^2$ specifications use the lender–industry–loan-type panel; HHI uses the lender–loan-type panel. Standard errors clustered at the lender level.



(a) Charge-off Rate

(b) Spread

Figure IA.6: Event-Study Plots: Loan-Outcome DID

Notes: Figures plot TWFE event-study coefficients with the single $Treated \times Post$ interaction replaced by quarter-by-treatment indicators ($t = -8$ to $t = 8$; endpoints binned; $t = -1$ as reference). Shaded areas represent 95% confidence intervals. The dashed vertical line marks the policy date. Standard errors clustered at the lender–industry–loan-type level ($N = 292{,}794$; 9,419 clusters).

G Placebo Timing Tests

Table IA.5 reports placebo timing tests for the extensive-margin specialization DID. The tests restrict the sample to pre-policy quarters only and assign a fictitious reform date in 2012:Q3 or 2013:Q3. If the main treatment effect reflects a genuine policy response rather than a pre-existing trend, these placebo DID coefficients should be statistically insignificant and small in magnitude.

Table IA.5: Placebo Timing Tests: Extensive-Margin DID

	Placebo: 2012:Q3		Placebo: 2013:Q3	
	$\lvert Spec \rvert$	$Spec^2$	$\lvert Spec \rvert$	$Spec^2$
Placebo $Treated \times Post$	-0.00112^*	-0.00099^*	-0.00106	-0.00122
	(0.00059)	(0.00056)	(0.00065)	(0.00088)
p-value	$[0.057]$	$[0.077]$	$[0.101]$	$[0.162]$
Observations	Pre-period only		Pre-period only	

Notes: Each column reports the placebo $Treated \times Post$ coefficient estimated on pre-policy quarters only, with the reform date moved to 2012:Q3 or 2013:Q3. Lender-industry-loan-type and quarter fixed effects included. Standard errors clustered at the lender level. None of the placebo coefficients is significant at the 5% level, supporting the parallel-trends assumption.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

H Honest Parallel-Trends Sensitivity

Table IA.6 reports the Roth (2022) honest confidence intervals for the loan-outcome DID. The method constructs confidence intervals that are robust to pre-trend violations of magnitude up to $\bar{M}$ times the largest pre-treatment coefficient change, under the relative-magnitudes restriction . The "Original" row reports the conventional 95% CI from the event-study regression.

Table IA.6: Roth (2022) Honest Confidence Intervals: Loan-Outcome DID

	Charge-off	Spread
Original 95% CI	$[-0.013,\ 0.028]$	$[-0.203,\ -0.020]$
Honest CI ($\bar{M} = 1$)	$[-0.036,\ 0.039]$	$[-0.272,\ 0.034]$
Honest CI ($\bar{M} = 2$)	$[-0.064,\ 0.065]$	$[-0.376,\ 0.123]$

Notes: Table reports honest confidence intervals following Roth (2022) for the loan-outcome event study. The method allows for pre-trend violations of magnitude up to $\bar{M}$ times the maximal pre-treatment coefficient change under the relative-magnitudes restriction. The event study uses 7 pre-treatment and 9 post-treatment periods. For charge-off rates, honest CIs remain centered near zero across both $\bar{M}$ values. For spreads, the honest CIs include zero at $\bar{M} \geq 1$, indicating that the spread result is sensitive to departures from exact parallel trends. Standard errors clustered at the lender–industry level.